

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                     |
|----------------------|---------------------|
| United States Patent | 12396667            |
| Kind Code            | B2                  |
| Date of Patent       | August 26, 2025     |
| Inventor(s)          | Dyell; David et al. |

---

### Alarm fatigue management systems and methods

---

#### Abstract

An apparatus for modifying alarms at a medical device for alarm fatigue management is provided and includes an alarm monitor, an alarm filter, an alarm modifier, a memory element for storing data, and a processor that executes instructions associated with the data, wherein the processor and the memory element cooperate such that the apparatus is configured for receiving an alarm condition from an alarm management engine, the alarm condition based on an alarm fatigue level of a user of the medical device, the alarm fatigue level based on at least a user fatigue model, a medical device model and a patient condition, receiving an alarm from the medical device, modifying the alarm according to the alarm condition, the alarm condition being configured to increase a likelihood of the user responding to the modified alarm, and generating an alarm indicator based on the modification.

---

**Inventors:** Dyell; David (Panama City, FL), Rogowski; Christopher (Newtown Square, PA), Kahler; Scott (Tuscaloosa, AL), Milan; Jennifer (Westminster, CO)

**Applicant:** VCCB Holdings, Inc. (Wilmington, DE)

**Family ID:** 1000008780621

**Assignee:** VCCB Holdings, Inc. (Wilmington, DE)

**Appl. No.:** 18/323187

**Filed:** May 24, 2023

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20230363678 A1   | Nov. 16, 2023    |

#### Related U.S. Application Data

continuation parent-doc US 17028822 20200922 US 11696712 child-doc US 18323187  
continuation parent-doc US 16735476 20200106 US 10813580 20201027 child-doc US 17028822

**Publication Classification**

**Int. Cl.:**    **A61B5/00** (20060101); **A61B5/16** (20060101); **G16H40/63** (20180101)

**U.S. Cl.:**

**CPC**           **A61B5/162** (20130101); **A61B5/746** (20130101); **G16H40/63** (20180101);

**Field of Classification Search**

**CPC:**        A61B (5/162); A61B (5/746); G16H (40/63)

---

**References Cited**

**U.S. PATENT DOCUMENTS**

| <b>Patent No.</b> | <b>Issued Date</b> | <b>Patentee Name</b>        | <b>U.S. Cl.</b> | <b>CPC</b> |
|-------------------|--------------------|-----------------------------|-----------------|------------|
| 4960128           | 12/1989            | Gordon et al.               | N/A             | N/A        |
| 4964408           | 12/1989            | Hink et al.                 | N/A             | N/A        |
| 5319355           | 12/1993            | Russek                      | N/A             | N/A        |
| 5337744           | 12/1993            | Branigan                    | N/A             | N/A        |
| 5341805           | 12/1993            | Stavridi et al.             | N/A             | N/A        |
| D353195           | 12/1993            | Savage et al.               | N/A             | N/A        |
| D353196           | 12/1993            | Savage et al.               | N/A             | N/A        |
| 5377676           | 12/1994            | Vari et al.                 | N/A             | N/A        |
| D359546           | 12/1994            | Savage et al.               | N/A             | N/A        |
| 5431170           | 12/1994            | Mathews                     | N/A             | N/A        |
| 5436499           | 12/1994            | Namavar et al.              | N/A             | N/A        |
| D361840           | 12/1994            | Savage et al.               | N/A             | N/A        |
| D362063           | 12/1994            | Savage et al.               | N/A             | N/A        |
| D363120           | 12/1994            | Savage et al.               | N/A             | N/A        |
| 5456252           | 12/1994            | Vari et al.                 | N/A             | N/A        |
| 5479934           | 12/1995            | Imran                       | N/A             | N/A        |
| 5482036           | 12/1995            | Diab et al.                 | N/A             | N/A        |
| 5494043           | 12/1995            | O'Sullivan et al.           | N/A             | N/A        |
| 5533511           | 12/1995            | Kaspari et al.              | N/A             | N/A        |
| 5561275           | 12/1995            | Savage et al.               | N/A             | N/A        |
| 5590649           | 12/1996            | Caro et al.                 | N/A             | N/A        |
| 5602924           | 12/1996            | Durand et al.               | N/A             | N/A        |
| 5638816           | 12/1996            | Kiani-Azarbayjany<br>et al. | N/A             | N/A        |
| 5638818           | 12/1996            | Diab et al.                 | N/A             | N/A        |
| 5645440           | 12/1996            | Tobler et al.               | N/A             | N/A        |
| 5671914           | 12/1996            | Kalkhoran et al.            | N/A             | N/A        |
| 5726440           | 12/1997            | Kalkhoran et al.            | N/A             | N/A        |
| D393830           | 12/1997            | Tobler et al.               | N/A             | N/A        |

|         |         |                     |     |     |
|---------|---------|---------------------|-----|-----|
| 5743262 | 12/1997 | Lepper, Jr. et al.  | N/A | N/A |
| 5747806 | 12/1997 | Khalil et al.       | N/A | N/A |
| 5750994 | 12/1997 | Schlager            | N/A | N/A |
| 5758644 | 12/1997 | Diab et al.         | N/A | N/A |
| 5760910 | 12/1997 | Lepper, Jr. et al.  | N/A | N/A |
| 5890929 | 12/1998 | Mills et al.        | N/A | N/A |
| 5919134 | 12/1998 | Diab                | N/A | N/A |
| 5987343 | 12/1998 | Kinast              | N/A | N/A |
| 5997343 | 12/1998 | Mills et al.        | N/A | N/A |
| 6002952 | 12/1998 | Diab et al.         | N/A | N/A |
| 6010937 | 12/1999 | Karam et al.        | N/A | N/A |
| 6027452 | 12/1999 | Flaherty et al.     | N/A | N/A |
| 6040578 | 12/1999 | Malin et al.        | N/A | N/A |
| 6066204 | 12/1999 | Haven               | N/A | N/A |
| 6115673 | 12/1999 | Malin et al.        | N/A | N/A |
| 6124597 | 12/1999 | Shehada et al.      | N/A | N/A |
| 6128521 | 12/1999 | Marro et al.        | N/A | N/A |
| 6129675 | 12/1999 | Jay                 | N/A | N/A |
| 6144868 | 12/1999 | Parker              | N/A | N/A |
| 6152754 | 12/1999 | Gerhardt et al.     | N/A | N/A |
| 6184521 | 12/2000 | Coffin, IV et al.   | N/A | N/A |
| 6232609 | 12/2000 | Snyder et al.       | N/A | N/A |
| 6241683 | 12/2000 | Macklem et al.      | N/A | N/A |
| 6253097 | 12/2000 | Aronow et al.       | N/A | N/A |
| 6255708 | 12/2000 | Sudharsanan et al.  | N/A | N/A |
| 6280381 | 12/2000 | Malin et al.        | N/A | N/A |
| 6285896 | 12/2000 | Tobler et al.       | N/A | N/A |
| 6308089 | 12/2000 | von der Ruhr et al. | N/A | N/A |
| 6317627 | 12/2000 | Ennen et al.        | N/A | N/A |
| 6321100 | 12/2000 | Parker              | N/A | N/A |
| 6334065 | 12/2000 | Al-Ali et al.       | N/A | N/A |
| 6360114 | 12/2001 | Diab et al.         | N/A | N/A |
| 6368283 | 12/2001 | Xu et al.           | N/A | N/A |
| 6411373 | 12/2001 | Garside et al.      | N/A | N/A |
| 6415167 | 12/2001 | Blank et al.        | N/A | N/A |
| 6430437 | 12/2001 | Marro               | N/A | N/A |
| 6430525 | 12/2001 | Weber et al.        | N/A | N/A |
| 6463311 | 12/2001 | Diab                | N/A | N/A |
| 6470199 | 12/2001 | Kopotic et al.      | N/A | N/A |
| 6487429 | 12/2001 | Hockersmith et al.  | N/A | N/A |
| 6505059 | 12/2002 | Kollias et al.      | N/A | N/A |
| 6525386 | 12/2002 | Mills et al.        | N/A | N/A |
| 6526300 | 12/2002 | Kiani et al.        | N/A | N/A |
| 6534012 | 12/2002 | Hazen et al.        | N/A | N/A |
| 6542764 | 12/2002 | Al-Ali et al.       | N/A | N/A |
| 6580086 | 12/2002 | Schulz et al.       | N/A | N/A |
| 6584336 | 12/2002 | Ali et al.          | N/A | N/A |
| 6585645 | 12/2002 | Hutchinson          | N/A | N/A |
| 6587196 | 12/2002 | Stippick et al.     | N/A | N/A |
| 6587199 | 12/2002 | Luu                 | N/A | N/A |

|         |         |                       |     |     |
|---------|---------|-----------------------|-----|-----|
| 6595316 | 12/2002 | Cybulski et al.       | N/A | N/A |
| 6597932 | 12/2002 | Tian et al.           | N/A | N/A |
| 6606511 | 12/2002 | Ali et al.            | N/A | N/A |
| 6635559 | 12/2002 | Greenwald et al.      | N/A | N/A |
| 6639668 | 12/2002 | Trepagnier            | N/A | N/A |
| 6640116 | 12/2002 | Diab                  | N/A | N/A |
| 6640117 | 12/2002 | Makarewicz et al.     | N/A | N/A |
| 6658276 | 12/2002 | Kiani et al.          | N/A | N/A |
| 6661161 | 12/2002 | Lanzo et al.          | N/A | N/A |
| 6681003 | 12/2003 | Linder et al.         | N/A | N/A |
| 6697656 | 12/2003 | Al-Ali                | N/A | N/A |
| 6697658 | 12/2003 | Al-Ali                | N/A | N/A |
| RE38476 | 12/2003 | Diab et al.           | N/A | N/A |
| RE38492 | 12/2003 | Diab et al.           | N/A | N/A |
| 6738652 | 12/2003 | Mattu et al.          | N/A | N/A |
| 6738671 | 12/2003 | Christophersom et al. | N/A | N/A |
| 6760607 | 12/2003 | Al-Ali                | N/A | N/A |
| 6788965 | 12/2003 | Ruchti et al.         | N/A | N/A |
| 6816241 | 12/2003 | Grubisic              | N/A | N/A |
| 6822564 | 12/2003 | Al-Ali                | N/A | N/A |
| 6850787 | 12/2004 | Weber et al.          | N/A | N/A |
| 6850788 | 12/2004 | Al-Ali                | N/A | N/A |
| 6876931 | 12/2004 | Lorenz et al.         | N/A | N/A |
| 6882883 | 12/2004 | Condie et al.         | N/A | N/A |
| 6920345 | 12/2004 | Al-Ali et al.         | N/A | N/A |
| 6934570 | 12/2004 | Kiani et al.          | N/A | N/A |
| 6943348 | 12/2004 | Coffin IV             | N/A | N/A |
| 6956649 | 12/2004 | Acosta et al.         | N/A | N/A |
| 6957107 | 12/2004 | Rogers et al.         | N/A | N/A |
| 6961598 | 12/2004 | Diab                  | N/A | N/A |
| 6970792 | 12/2004 | Diab                  | N/A | N/A |
| 6985764 | 12/2005 | Mason et al.          | N/A | N/A |
| 6990364 | 12/2005 | Ruchti et al.         | N/A | N/A |
| 6998247 | 12/2005 | Monfre et al.         | N/A | N/A |
| 7003338 | 12/2005 | Weber et al.          | N/A | N/A |
| 7015451 | 12/2005 | Dalke et al.          | N/A | N/A |
| 7027849 | 12/2005 | Al-Ali                | N/A | N/A |
| 7027872 | 12/2005 | Thompson              | N/A | N/A |
| 7077806 | 12/2005 | Ackermann et al.      | N/A | N/A |
| D526719 | 12/2005 | Richie, Jr. et al.    | N/A | N/A |
| 7096052 | 12/2005 | Mason et al.          | N/A | N/A |
| 7096054 | 12/2005 | Abdul-Hafiz et al.    | N/A | N/A |
| D529616 | 12/2005 | Deros et al.          | N/A | N/A |
| 7133710 | 12/2005 | Acosta et al.         | N/A | N/A |
| 7142901 | 12/2005 | Kiani et al.          | N/A | N/A |
| 7156809 | 12/2006 | Quy                   | N/A | N/A |
| 7181283 | 12/2006 | Hettrick et al.       | N/A | N/A |
| 7225006 | 12/2006 | Al-Ali et al.         | N/A | N/A |
| RE39672 | 12/2006 | Shehada et al.        | N/A | N/A |

|         |         |                  |     |     |
|---------|---------|------------------|-----|-----|
| 7229409 | 12/2006 | Ito et al.       | N/A | N/A |
| 7248916 | 12/2006 | Bardy            | N/A | N/A |
| 7254429 | 12/2006 | Schurman et al.  | N/A | N/A |
| 7254431 | 12/2006 | Al-Ali et al.    | N/A | N/A |
| 7254434 | 12/2006 | Schulz et al.    | N/A | N/A |
| 7274955 | 12/2006 | Kiani et al.     | N/A | N/A |
| D554263 | 12/2006 | Al-Ali et al.    | N/A | N/A |
| 7280858 | 12/2006 | Al-Ali et al.    | N/A | N/A |
| 7289835 | 12/2006 | Mansfield et al. | N/A | N/A |
| 7292883 | 12/2006 | De Felice et al. | N/A | N/A |
| 7341559 | 12/2007 | Schulz et al.    | N/A | N/A |
| 7343186 | 12/2007 | Lamego et al.    | N/A | N/A |
| D566282 | 12/2007 | Al-Ali et al.    | N/A | N/A |
| 7356365 | 12/2007 | Schurman         | N/A | N/A |
| 7371981 | 12/2007 | Abdul-Hafiz      | N/A | N/A |
| 7373193 | 12/2007 | Al-Ali et al.    | N/A | N/A |
| 7377794 | 12/2007 | Al-Ali et al.    | N/A | N/A |
| 7395158 | 12/2007 | Monfre et al.    | N/A | N/A |
| 7415297 | 12/2007 | Al-Ali et al.    | N/A | N/A |
| 7438683 | 12/2007 | Al-Ali et al.    | N/A | N/A |
| 7483729 | 12/2008 | Al-Ali et al.    | N/A | N/A |
| D587657 | 12/2008 | Al-Ali et al.    | N/A | N/A |
| 7500950 | 12/2008 | Al-Ali et al.    | N/A | N/A |
| 7509494 | 12/2008 | Al-Ali           | N/A | N/A |
| 7510849 | 12/2008 | Schurman et al.  | N/A | N/A |
| 7514725 | 12/2008 | Wojtczuk et al.  | N/A | N/A |
| 7519406 | 12/2008 | Blank et al.     | N/A | N/A |
| D592507 | 12/2008 | Wachman et al.   | N/A | N/A |
| 7530942 | 12/2008 | Diab             | N/A | N/A |
| 7593230 | 12/2008 | Abul-Haj et al.  | N/A | N/A |
| 7596398 | 12/2008 | Al-Ali et al.    | N/A | N/A |
| 7606608 | 12/2008 | Blank et al.     | N/A | N/A |
| 7620674 | 12/2008 | Ruchti et al.    | N/A | N/A |
| D606659 | 12/2008 | Kiani et al.     | N/A | N/A |
| 7629039 | 12/2008 | Eckerbom et al.  | N/A | N/A |
| 7640140 | 12/2008 | Ruchti et al.    | N/A | N/A |
| 7647083 | 12/2009 | Al-Ali et al.    | N/A | N/A |
| D609193 | 12/2009 | Al-Ali et al.    | N/A | N/A |
| D614305 | 12/2009 | Al-Ali et al.    | N/A | N/A |
| 7697966 | 12/2009 | Monfre et al.    | N/A | N/A |
| 7698105 | 12/2009 | Ruchti et al.    | N/A | N/A |
| RE41317 | 12/2009 | Parker           | N/A | N/A |
| RE41333 | 12/2009 | Blank et al.     | N/A | N/A |
| 7715919 | 12/2009 | Osorio et al.    | N/A | N/A |
| 7729733 | 12/2009 | Al-Ali et al.    | N/A | N/A |
| 7761127 | 12/2009 | Al-Ali et al.    | N/A | N/A |
| 7764982 | 12/2009 | Dalke et al.     | N/A | N/A |
| D621516 | 12/2009 | Kiani et al.     | N/A | N/A |
| 7791155 | 12/2009 | Diab             | N/A | N/A |
| 7801612 | 12/2009 | Johnson et al.   | N/A | N/A |

|         |         |                          |     |     |
|---------|---------|--------------------------|-----|-----|
| RE41912 | 12/2009 | Parker                   | N/A | N/A |
| 7880626 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 7909772 | 12/2010 | Popov et al.             | N/A | N/A |
| 7919713 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 7937128 | 12/2010 | Al-Ali                   | N/A | N/A |
| 7937129 | 12/2010 | Mason et al.             | N/A | N/A |
| 7941199 | 12/2010 | Kiani                    | N/A | N/A |
| 7957780 | 12/2010 | Lamego et al.            | N/A | N/A |
| 7962188 | 12/2010 | Kiani et al.             | N/A | N/A |
| 7967749 | 12/2010 | Hutchinson et al.        | N/A | N/A |
| 7976472 | 12/2010 | Kiani                    | N/A | N/A |
| 7990382 | 12/2010 | Kiani                    | N/A | N/A |
| 8002701 | 12/2010 | John et al.              | N/A | N/A |
| 8008088 | 12/2010 | Bellott et al.           | N/A | N/A |
| RE42753 | 12/2010 | Kiani-Azarbayjany et al. | N/A | N/A |
| 8028701 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 8048040 | 12/2010 | Kiani                    | N/A | N/A |
| 8050728 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| RE43169 | 12/2011 | Parker                   | N/A | N/A |
| 8118620 | 12/2011 | Al-Ali et al.            | N/A | N/A |
| 8123720 | 12/2011 | Solomon                  | N/A | N/A |
| 8126735 | 12/2011 | Dicks et al.             | N/A | N/A |
| 8130105 | 12/2011 | Al-Ali et al.            | N/A | N/A |
| 8165893 | 12/2011 | Goldberg et al.          | N/A | N/A |
| 8182443 | 12/2011 | Kiani                    | N/A | N/A |
| 8183451 | 12/2011 | Panaiotis                | N/A | N/A |
| 8190223 | 12/2011 | Al-Ali et al.            | N/A | N/A |
| 8203438 | 12/2011 | Kiani et al.             | N/A | N/A |
| 8203704 | 12/2011 | Merritt et al.           | N/A | N/A |
| 8219172 | 12/2011 | Schurman et al.          | N/A | N/A |
| 8224411 | 12/2011 | Al-Ali et al.            | N/A | N/A |
| 8229532 | 12/2011 | Davis                    | N/A | N/A |
| 8233955 | 12/2011 | Al-Ali et al.            | N/A | N/A |
| 8255026 | 12/2011 | Al-Ali                   | N/A | N/A |
| 8255240 | 12/2011 | O'Hanlon et al.          | N/A | N/A |
| 8265723 | 12/2011 | McHale et al.            | N/A | N/A |
| 8274360 | 12/2011 | Sampath et al.           | N/A | N/A |
| 8280473 | 12/2011 | Al-Ali                   | N/A | N/A |
| 8310374 | 12/2011 | Grubis et al.            | N/A | N/A |
| 8315683 | 12/2011 | Al-Ali et al.            | N/A | N/A |
| RE43860 | 12/2011 | Parker                   | N/A | N/A |
| 8346330 | 12/2012 | Lamego                   | N/A | N/A |
| 8353842 | 12/2012 | Al-Ali et al.            | N/A | N/A |
| 8355766 | 12/2012 | MacNeish, III et al.     | N/A | N/A |
| 8374665 | 12/2012 | Lamego                   | N/A | N/A |
| 8388353 | 12/2012 | Kiani et al.             | N/A | N/A |
| 8395498 | 12/2012 | Gaskill et al.           | N/A | N/A |
| 8401602 | 12/2012 | Kiani                    | N/A | N/A |
| 8414499 | 12/2012 | Al-Ali et al.            | N/A | N/A |

|         |         |                      |     |     |
|---------|---------|----------------------|-----|-----|
| 8417662 | 12/2012 | Gawlick              | N/A | N/A |
| 8418524 | 12/2012 | Al-Ali               | N/A | N/A |
| 8425414 | 12/2012 | Eveland              | N/A | N/A |
| 8428967 | 12/2012 | Olsen et al.         | N/A | N/A |
| 8430817 | 12/2012 | Al-Ali et al.        | N/A | N/A |
| 8437825 | 12/2012 | Dalvi et al.         | N/A | N/A |
| 8438038 | 12/2012 | Cosentino et al.     | N/A | N/A |
| 8451113 | 12/2012 | Mazar et al.         | N/A | N/A |
| 8455290 | 12/2012 | Siskavich            | N/A | N/A |
| 8457707 | 12/2012 | Kiani                | N/A | N/A |
| 8471713 | 12/2012 | Poeze et al.         | N/A | N/A |
| 8473020 | 12/2012 | Kiani et al.         | N/A | N/A |
| 8487758 | 12/2012 | Istoc                | N/A | N/A |
| 8509867 | 12/2012 | Workman et al.       | N/A | N/A |
| 8515509 | 12/2012 | Bruinsma et al.      | N/A | N/A |
| 8523781 | 12/2012 | Al-Ali               | N/A | N/A |
| D692145 | 12/2012 | Al-Ali et al.        | N/A | N/A |
| 8571617 | 12/2012 | Reichgott et al.     | N/A | N/A |
| 8571618 | 12/2012 | Lamego et al.        | N/A | N/A |
| 8571619 | 12/2012 | Al-Ali et al.        | N/A | N/A |
| 8577431 | 12/2012 | Lamego et al.        | N/A | N/A |
| 8584345 | 12/2012 | Al-Ali et al.        | N/A | N/A |
| 8588880 | 12/2012 | Abdul-Hafiz et al.   | N/A | N/A |
| 8630691 | 12/2013 | Lamego et al.        | N/A | N/A |
| 8641631 | 12/2013 | Sierra et al.        | N/A | N/A |
| 8652060 | 12/2013 | Al-Ali               | N/A | N/A |
| 8666468 | 12/2013 | Al-Ali               | N/A | N/A |
| 8670811 | 12/2013 | O'Reilly             | N/A | N/A |
| RE44823 | 12/2013 | Parker               | N/A | N/A |
| RE44875 | 12/2013 | Kiani et al.         | N/A | N/A |
| 8688183 | 12/2013 | Bruinsma et al.      | N/A | N/A |
| 8690799 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8702627 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8712494 | 12/2013 | MacNeish, III et al. | N/A | N/A |
| 8715206 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8718949 | 12/2013 | Blomquist et al.     | N/A | N/A |
| 8723677 | 12/2013 | Kiani                | N/A | N/A |
| 8740792 | 12/2013 | Kiani et al.         | N/A | N/A |
| 8755535 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8755872 | 12/2013 | Marinow              | N/A | N/A |
| 8764671 | 12/2013 | Kiani                | N/A | N/A |
| 8768423 | 12/2013 | Shakespeare et al.   | N/A | N/A |
| 8771204 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8781544 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8790268 | 12/2013 | Al-Ali               | N/A | N/A |
| 8801613 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8821397 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8821415 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8830449 | 12/2013 | Lamego et al.        | N/A | N/A |
| 8840549 | 12/2013 | Al-Ali et al.        | N/A | N/A |

|         |         |                  |     |     |
|---------|---------|------------------|-----|-----|
| 8852094 | 12/2013 | Al-Ali et al.    | N/A | N/A |
| 8852994 | 12/2013 | Wojtczuk et al.  | N/A | N/A |
| 8897847 | 12/2013 | Al-Ali           | N/A | N/A |
| 8911377 | 12/2013 | Al-Ali           | N/A | N/A |
| 8989831 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 8998809 | 12/2014 | Kiani            | N/A | N/A |
| 9066666 | 12/2014 | Kiani            | N/A | N/A |
| 9066680 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9095316 | 12/2014 | Welch et al.     | N/A | N/A |
| 9106038 | 12/2014 | Telfort et al.   | N/A | N/A |
| 9107625 | 12/2014 | Telfort et al.   | N/A | N/A |
| 9131881 | 12/2014 | Diab et al.      | N/A | N/A |
| 9138180 | 12/2014 | Coverston et al. | N/A | N/A |
| 9153112 | 12/2014 | Kiani et al.     | N/A | N/A |
| 9192329 | 12/2014 | Al-Ali           | N/A | N/A |
| 9192351 | 12/2014 | Telfort et al.   | N/A | N/A |
| 9195385 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9211095 | 12/2014 | Al-Ali           | N/A | N/A |
| 9218454 | 12/2014 | Kiani et al.     | N/A | N/A |
| 9245668 | 12/2015 | Vo et al.        | N/A | N/A |
| 9267572 | 12/2015 | Barker et al.    | N/A | N/A |
| 9277880 | 12/2015 | Poeze et al.     | N/A | N/A |
| 9307928 | 12/2015 | Al-Ali et al.    | N/A | N/A |
| 9323894 | 12/2015 | Kiani            | N/A | N/A |
| D755392 | 12/2015 | Hwang et al.     | N/A | N/A |
| 9326712 | 12/2015 | Kiani            | N/A | N/A |
| 9392945 | 12/2015 | Al-Ali et al.    | N/A | N/A |
| 9408542 | 12/2015 | Kinast et al.    | N/A | N/A |
| 9436645 | 12/2015 | Al-Ali et al.    | N/A | N/A |
| 9445759 | 12/2015 | Lamego et al.    | N/A | N/A |
| 9474474 | 12/2015 | Lamego et al.    | N/A | N/A |
| 9480435 | 12/2015 | Olsen            | N/A | N/A |
| 9510779 | 12/2015 | Poeze et al.     | N/A | N/A |
| 9517024 | 12/2015 | Kiani et al.     | N/A | N/A |
| 9532722 | 12/2016 | Lamego et al.    | N/A | N/A |
| 9560996 | 12/2016 | Kiani            | N/A | N/A |
| 9579039 | 12/2016 | Jansen et al.    | N/A | N/A |
| 9582988 | 12/2016 | Gross            | N/A | N/A |
| 9622692 | 12/2016 | Lamego et al.    | N/A | N/A |
| D788312 | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 9649054 | 12/2016 | Lamego et al.    | N/A | N/A |
| 9697928 | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 9717458 | 12/2016 | Lamego et al.    | N/A | N/A |
| 9724016 | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 9724024 | 12/2016 | Al-Ali           | N/A | N/A |
| 9724025 | 12/2016 | Kiani et al.     | N/A | N/A |
| 9749232 | 12/2016 | Sampath et al.   | N/A | N/A |
| 9750442 | 12/2016 | Olsen            | N/A | N/A |
| 9750461 | 12/2016 | Telfort          | N/A | N/A |
| 9775545 | 12/2016 | Al-Ali et al.    | N/A | N/A |

|          |         |                 |     |     |
|----------|---------|-----------------|-----|-----|
| 9778079  | 12/2016 | Al-Ali et al.   | N/A | N/A |
| 9782077  | 12/2016 | Lamego et al.   | N/A | N/A |
| 9787568  | 12/2016 | Lamego et al.   | N/A | N/A |
| 9808188  | 12/2016 | Perea et al.    | N/A | N/A |
| 9839379  | 12/2016 | Al-Ali et al.   | N/A | N/A |
| 9839381  | 12/2016 | Weber et al.    | N/A | N/A |
| 9847749  | 12/2016 | Kiani et al.    | N/A | N/A |
| 9848800  | 12/2016 | Lee et al.      | N/A | N/A |
| 9861298  | 12/2017 | Eckerbom et al. | N/A | N/A |
| 9861305  | 12/2017 | Weber et al.    | N/A | N/A |
| 9877650  | 12/2017 | Muhsin et al.   | N/A | N/A |
| 9891079  | 12/2017 | Dalvi           | N/A | N/A |
| 9924897  | 12/2017 | Abdul-Hafiz     | N/A | N/A |
| 9936917  | 12/2017 | Poeze et al.    | N/A | N/A |
| 9955937  | 12/2017 | Telfort         | N/A | N/A |
| 9965946  | 12/2017 | Al-Ali et al.   | N/A | N/A |
| D820865  | 12/2017 | Muhsin et al.   | N/A | N/A |
| 9986952  | 12/2017 | Dalvi et al.    | N/A | N/A |
| D822215  | 12/2017 | Al-Ali et al.   | N/A | N/A |
| D822216  | 12/2017 | Barker et al.   | N/A | N/A |
| 10010276 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 10055974 | 12/2017 | Wilson          | N/A | N/A |
| 10086138 | 12/2017 | Novak, Jr.      | N/A | N/A |
| 10111591 | 12/2017 | Dyell et al.    | N/A | N/A |
| D833624  | 12/2017 | DeJong et al.   | N/A | N/A |
| 10123729 | 12/2017 | Dyell et al.    | N/A | N/A |
| D835282  | 12/2017 | Barker et al.   | N/A | N/A |
| D835283  | 12/2017 | Barker et al.   | N/A | N/A |
| D835284  | 12/2017 | Barker et al.   | N/A | N/A |
| D835285  | 12/2017 | Barker et al.   | N/A | N/A |
| 10149616 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 10154815 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 10159412 | 12/2017 | Lamego et al.   | N/A | N/A |
| 10188348 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| RE47218  | 12/2018 | Al-Ali          | N/A | N/A |
| RE47244  | 12/2018 | Kiani et al.    | N/A | N/A |
| RE47249  | 12/2018 | Kiani et al.    | N/A | N/A |
| 10205291 | 12/2018 | Scruggs et al.  | N/A | N/A |
| 10226187 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10231657 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10231670 | 12/2018 | Blank et al.    | N/A | N/A |
| RE47353  | 12/2018 | Kiani et al.    | N/A | N/A |
| 10279247 | 12/2018 | Kiani           | N/A | N/A |
| 10292664 | 12/2018 | Al-Ali          | N/A | N/A |
| 10299720 | 12/2018 | Brown et al.    | N/A | N/A |
| 10327337 | 12/2018 | Schmidt et al.  | N/A | N/A |
| 10327713 | 12/2018 | Barker et al.   | N/A | N/A |
| 10332630 | 12/2018 | Al-Ali          | N/A | N/A |
| 10383520 | 12/2018 | Wojtczuk et al. | N/A | N/A |
| 10383527 | 12/2018 | Al-Ali          | N/A | N/A |

|          |         |                    |     |     |
|----------|---------|--------------------|-----|-----|
| 10388120 | 12/2018 | Muhsin et al.      | N/A | N/A |
| D864120  | 12/2018 | Forrest et al.     | N/A | N/A |
| 10441181 | 12/2018 | Telfort et al.     | N/A | N/A |
| 10441196 | 12/2018 | Eckerbom et al.    | N/A | N/A |
| 10448844 | 12/2018 | Al-Ali et al.      | N/A | N/A |
| 10448871 | 12/2018 | Al-Ali et al.      | N/A | N/A |
| 10456038 | 12/2018 | Lamego et al.      | N/A | N/A |
| 10463340 | 12/2018 | Telfort et al.     | N/A | N/A |
| 10471159 | 12/2018 | Lapotko et al.     | N/A | N/A |
| 10505311 | 12/2018 | Al-Ali et al.      | N/A | N/A |
| 10524738 | 12/2019 | Olsen              | N/A | N/A |
| 10532174 | 12/2019 | Al-Ali             | N/A | N/A |
| 10537285 | 12/2019 | Shreim et al.      | N/A | N/A |
| 10542903 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| 10555678 | 12/2019 | Dalvi et al.       | N/A | N/A |
| 10568553 | 12/2019 | O'Neil et al.      | N/A | N/A |
| RE47882  | 12/2019 | Al-Ali             | N/A | N/A |
| 10608817 | 12/2019 | Haider et al.      | N/A | N/A |
| D880477  | 12/2019 | Forrest et al.     | N/A | N/A |
| 10617302 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| 10617335 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| 10637181 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| D886849  | 12/2019 | Muhsin et al.      | N/A | N/A |
| D887548  | 12/2019 | Abdul-Hafiz et al. | N/A | N/A |
| D887549  | 12/2019 | Abdul-Hafiz et al. | N/A | N/A |
| 10667764 | 12/2019 | Ahmed et al.       | N/A | N/A |
| D890708  | 12/2019 | Forrest et al.     | N/A | N/A |
| 10721785 | 12/2019 | Al-Ali             | N/A | N/A |
| 10736518 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| 10750984 | 12/2019 | Pauley et al.      | N/A | N/A |
| D897098  | 12/2019 | Al-Ali             | N/A | N/A |
| 10779098 | 12/2019 | Iswanto et al.     | N/A | N/A |
| 10813580 | 12/2019 | Dyell et al.       | N/A | N/A |
| 10827961 | 12/2019 | Iyengar et al.     | N/A | N/A |
| 10828007 | 12/2019 | Telfort et al.     | N/A | N/A |
| 10832818 | 12/2019 | Muhsin et al.      | N/A | N/A |
| 10849554 | 12/2019 | Shreim et al.      | N/A | N/A |
| 10856750 | 12/2019 | Indorf             | N/A | N/A |
| D906970  | 12/2020 | Forrest et al.     | N/A | N/A |
| D908213  | 12/2020 | Abdul-Hafiz et al. | N/A | N/A |
| 10918281 | 12/2020 | Al-Ali et al.      | N/A | N/A |
| 10932705 | 12/2020 | Muhsin et al.      | N/A | N/A |
| 10932729 | 12/2020 | Kiani et al.       | N/A | N/A |
| 10939878 | 12/2020 | Kiani et al.       | N/A | N/A |
| 10956950 | 12/2020 | Al-Ali et al.      | N/A | N/A |
| D916135  | 12/2020 | Indorf et al.      | N/A | N/A |
| D917046  | 12/2020 | Abdul-Hafiz et al. | N/A | N/A |
| D917550  | 12/2020 | Indorf et al.      | N/A | N/A |
| D917564  | 12/2020 | Indorf et al.      | N/A | N/A |
| D917704  | 12/2020 | Al-Ali et al.      | N/A | N/A |

|          |         |                 |     |     |
|----------|---------|-----------------|-----|-----|
| 10987066 | 12/2020 | Chandran et al. | N/A | N/A |
| 10991135 | 12/2020 | Al-Ali et al.   | N/A | N/A |
| D919094  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| D919100  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11006867 | 12/2020 | Al-Ali          | N/A | N/A |
| D921202  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11024064 | 12/2020 | Muhsin et al.   | N/A | N/A |
| 11026604 | 12/2020 | Chen et al.     | N/A | N/A |
| D925597  | 12/2020 | Chandran et al. | N/A | N/A |
| D927699  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11076777 | 12/2020 | Lee et al.      | N/A | N/A |
| 11114188 | 12/2020 | Poeze et al.    | N/A | N/A |
| D933232  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| D933233  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| D933234  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11145408 | 12/2020 | Sampath et al.  | N/A | N/A |
| 11147518 | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11185262 | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11191484 | 12/2020 | Kiani et al.    | N/A | N/A |
| D946596  | 12/2021 | Ahmed           | N/A | N/A |
| D946597  | 12/2021 | Ahmed           | N/A | N/A |
| D946598  | 12/2021 | Ahmed           | N/A | N/A |
| D946617  | 12/2021 | Ahmed           | N/A | N/A |
| 11272839 | 12/2021 | Al-Ali et al.   | N/A | N/A |
| 11289199 | 12/2021 | Al-Ali          | N/A | N/A |
| RE49034  | 12/2021 | Al-Ali          | N/A | N/A |
| 11298021 | 12/2021 | Muhsin et al.   | N/A | N/A |
| D950580  | 12/2021 | Ahmed           | N/A | N/A |
| D950599  | 12/2021 | Ahmed           | N/A | N/A |
| D950738  | 12/2021 | Al-Ali et al.   | N/A | N/A |
| D957648  | 12/2021 | Al-Ali          | N/A | N/A |
| 11382567 | 12/2021 | O'Brien et al.  | N/A | N/A |
| 11389093 | 12/2021 | Triman et al.   | N/A | N/A |
| 11406286 | 12/2021 | Al-Ali et al.   | N/A | N/A |
| 11417426 | 12/2021 | Muhsin et al.   | N/A | N/A |
| 11439329 | 12/2021 | Lamego          | N/A | N/A |
| 11445948 | 12/2021 | Scruggs et al.  | N/A | N/A |
| D965789  | 12/2021 | Al-Ali et al.   | N/A | N/A |
| D967433  | 12/2021 | Al-Ali et al.   | N/A | N/A |
| 11464410 | 12/2021 | Muhsin          | N/A | N/A |
| 11504058 | 12/2021 | Sharma et al.   | N/A | N/A |
| 11504066 | 12/2021 | Dalvi et al.    | N/A | N/A |
| D971933  | 12/2021 | Ahmed           | N/A | N/A |
| D973072  | 12/2021 | Ahmed           | N/A | N/A |
| D973685  | 12/2021 | Ahmed           | N/A | N/A |
| D973686  | 12/2021 | Ahmed           | N/A | N/A |
| D974193  | 12/2022 | Forrest et al.  | N/A | N/A |
| D979516  | 12/2022 | Al-Ali et al.   | N/A | N/A |
| D980091  | 12/2022 | Forrest et al.  | N/A | N/A |
| 11596363 | 12/2022 | Lamego          | N/A | N/A |

|          |         |                      |     |     |
|----------|---------|----------------------|-----|-----|
| 11627919 | 12/2022 | Kiani et al.         | N/A | N/A |
| 11637437 | 12/2022 | Al-Ali et al.        | N/A | N/A |
| D985498  | 12/2022 | Al-Ali et al.        | N/A | N/A |
| 11653862 | 12/2022 | Dalvi et al.         | N/A | N/A |
| D989112  | 12/2022 | Muhsin et al.        | N/A | N/A |
| D989327  | 12/2022 | Al-Ali et al.        | N/A | N/A |
| 11678829 | 12/2022 | Al-Ali et al.        | N/A | N/A |
| 11679579 | 12/2022 | Al-Ali               | N/A | N/A |
| 11684296 | 12/2022 | Vo et al.            | N/A | N/A |
| 11692934 | 12/2022 | Normand et al.       | N/A | N/A |
| 11696712 | 12/2022 | Dyell et al.         | N/A | N/A |
| 11701043 | 12/2022 | Al-Ali et al.        | N/A | N/A |
| D997365  | 12/2022 | Hwang                | N/A | N/A |
| 11721105 | 12/2022 | Ranasinghe et al.    | N/A | N/A |
| 11730379 | 12/2022 | Ahmed et al.         | N/A | N/A |
| D998625  | 12/2022 | Indorf et al.        | N/A | N/A |
| D998630  | 12/2022 | Indorf et al.        | N/A | N/A |
| D998631  | 12/2022 | Indorf et al.        | N/A | N/A |
| D999244  | 12/2022 | Indorf et al.        | N/A | N/A |
| D999245  | 12/2022 | Indorf et al.        | N/A | N/A |
| D999246  | 12/2022 | Indorf et al.        | N/A | N/A |
| 11766198 | 12/2022 | Pauley et al.        | N/A | N/A |
| D1000975 | 12/2022 | Al-Ali et al.        | N/A | N/A |
| 11803623 | 12/2022 | Kiani et al.         | N/A | N/A |
| 11832940 | 12/2022 | Diab et al.          | N/A | N/A |
| D1013179 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 11872156 | 12/2023 | Telfort et al.       | N/A | N/A |
| 11879960 | 12/2023 | Ranasinghe et al.    | N/A | N/A |
| 11883129 | 12/2023 | Olsen                | N/A | N/A |
| D1022729 | 12/2023 | Forrest et al.       | N/A | N/A |
| 11951186 | 12/2023 | Krishnamani et al.   | N/A | N/A |
| 11974833 | 12/2023 | Forrest et al.       | N/A | N/A |
| 11986067 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 11986289 | 12/2023 | Dalvi et al.         | N/A | N/A |
| 11986305 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| D1031729 | 12/2023 | Forrest et al.       | N/A | N/A |
| 12004869 | 12/2023 | Kiani et al.         | N/A | N/A |
| 12014328 | 12/2023 | Wachman et al.       | N/A | N/A |
| D1036293 | 12/2023 | Al-Ali               | N/A | N/A |
| D1037462 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 12029844 | 12/2023 | Pauley et al.        | N/A | N/A |
| 12048534 | 12/2023 | Vo et al.            | N/A | N/A |
| 12064217 | 12/2023 | Ahmed et al.         | N/A | N/A |
| 12066426 | 12/2023 | Lapotko et al.       | N/A | N/A |
| D1041511 | 12/2023 | Indorf et al.        | N/A | N/A |
| D1042596 | 12/2023 | DeJong et al.        | N/A | N/A |
| D1042852 | 12/2023 | Hwang                | N/A | N/A |
| 12076159 | 12/2023 | Belur Nagaraj et al. | N/A | N/A |
| 12082926 | 12/2023 | Sharma et al.        | N/A | N/A |
| D1044828 | 12/2023 | Chandran et al.      | N/A | N/A |

|              |         |                     |         |                |
|--------------|---------|---------------------|---------|----------------|
| D1048571     | 12/2023 | Yu et al.           | N/A     | N/A            |
| D1048908     | 12/2023 | Al-Ali et al.       | N/A     | N/A            |
| 12106752     | 12/2023 | Campbell et al.     | N/A     | N/A            |
| 12114974     | 12/2023 | Al-Ali et al.       | N/A     | N/A            |
| 12126683     | 12/2023 | Koo et al.          | N/A     | N/A            |
| 12127838     | 12/2023 | Olsen et al.        | N/A     | N/A            |
| 12128213     | 12/2023 | Kiani et al.        | N/A     | N/A            |
| 12131661     | 12/2023 | Pauley et al.       | N/A     | N/A            |
| D1050910     | 12/2023 | Al-Ali et al.       | N/A     | N/A            |
| 12178572     | 12/2023 | Pauley et al.       | N/A     | N/A            |
| 12178581     | 12/2023 | Telfort et al.      | N/A     | N/A            |
| 12178852     | 12/2023 | Kiani et al.        | N/A     | N/A            |
| D1057159     | 12/2024 | DeJong et al.       | N/A     | N/A            |
| D1057160     | 12/2024 | DeJong et al.       | N/A     | N/A            |
| 12198790     | 12/2024 | Al-Ali              | N/A     | N/A            |
| 12200421     | 12/2024 | Campbell et al.     | N/A     | N/A            |
| 2001/0034477 | 12/2000 | Mansfield et al.    | N/A     | N/A            |
| 2001/0039483 | 12/2000 | Brand et al.        | N/A     | N/A            |
| 2002/0010401 | 12/2001 | Bushmakina et al.   | N/A     | N/A            |
| 2002/0058864 | 12/2001 | Mansfield et al.    | N/A     | N/A            |
| 2002/0133080 | 12/2001 | Apruzzese et al.    | N/A     | N/A            |
| 2003/0013975 | 12/2002 | Kiani               | N/A     | N/A            |
| 2003/0018243 | 12/2002 | Gerhardt et al.     | N/A     | N/A            |
| 2003/0144582 | 12/2002 | Cohen et al.        | N/A     | N/A            |
| 2003/0156288 | 12/2002 | Barnum et al.       | N/A     | N/A            |
| 2003/0212312 | 12/2002 | Coffin, IV et al.   | N/A     | N/A            |
| 2004/0103001 | 12/2003 | Mazar et al.        | N/A     | N/A            |
| 2004/0106163 | 12/2003 | Workman, Jr. et al. | N/A     | N/A            |
| 2005/0055242 | 12/2004 | Bello et al.        | N/A     | N/A            |
| 2005/0055276 | 12/2004 | Kiani et al.        | N/A     | N/A            |
| 2005/0101841 | 12/2004 | Kaylor et al.       | N/A     | N/A            |
| 2005/0234317 | 12/2004 | Kiani               | N/A     | N/A            |
| 2006/0064020 | 12/2005 | Burnes et al.       | N/A     | N/A            |
| 2006/0073719 | 12/2005 | Kiani               | N/A     | N/A            |
| 2006/0074722 | 12/2005 | Chu                 | N/A     | N/A            |
| 2006/0189871 | 12/2005 | Al-Ali et al.       | N/A     | N/A            |
| 2007/0073116 | 12/2006 | Kiani et al.        | N/A     | N/A            |
| 2007/0180140 | 12/2006 | Welch et al.        | N/A     | N/A            |
| 2007/0244377 | 12/2006 | Cozad et al.        | N/A     | N/A            |
| 2008/0027499 | 12/2007 | Srivathsa et al.    | N/A     | N/A            |
| 2008/0064965 | 12/2007 | Jay et al.          | N/A     | N/A            |
| 2008/0094228 | 12/2007 | Welch et al.        | N/A     | N/A            |
| 2008/0103375 | 12/2007 | Kiani               | N/A     | N/A            |
| 2008/0146895 | 12/2007 | Olson               | 600/301 | G16H<br>40/67  |
| 2008/0221418 | 12/2007 | Al-Ali et al.       | N/A     | N/A            |
| 2009/0036759 | 12/2008 | Ault et al.         | N/A     | N/A            |
| 2009/0048500 | 12/2008 | Corn                | 600/315 | A61B<br>5/4094 |
| 2009/0076342 | 12/2008 | Amurthur et al.     | N/A     | N/A            |

|              |         |                     |         |               |
|--------------|---------|---------------------|---------|---------------|
| 2009/0093687 | 12/2008 | Telfort et al.      | N/A     | N/A           |
| 2009/0095926 | 12/2008 | MacNeish, III       | N/A     | N/A           |
| 2009/0247984 | 12/2008 | Lamego et al.       | N/A     | N/A           |
| 2009/0275807 | 12/2008 | Sitzman et al.      | N/A     | N/A           |
| 2009/0275813 | 12/2008 | Davis               | N/A     | N/A           |
| 2009/0275844 | 12/2008 | Al-Ali              | N/A     | N/A           |
| 2009/0299162 | 12/2008 | Brauker et al.      | N/A     | N/A           |
| 2010/0004518 | 12/2009 | Vo et al.           | N/A     | N/A           |
| 2010/0030040 | 12/2009 | Poeze et al.        | N/A     | N/A           |
| 2010/0070188 | 12/2009 | Solomon             | N/A     | N/A           |
| 2010/0099964 | 12/2009 | O'Reilly et al.     | N/A     | N/A           |
| 2010/0234718 | 12/2009 | Sampath et al.      | N/A     | N/A           |
| 2010/0259395 | 12/2009 | Nuthi               | N/A     | N/A           |
| 2010/0270257 | 12/2009 | Wachman et al.      | N/A     | N/A           |
| 2011/0028806 | 12/2010 | Merritt et al.      | N/A     | N/A           |
| 2011/0028809 | 12/2010 | Goodman             | N/A     | N/A           |
| 2011/0040197 | 12/2010 | Welch et al.        | N/A     | N/A           |
| 2011/0077967 | 12/2010 | Kapu et al.         | N/A     | N/A           |
| 2011/0082711 | 12/2010 | Poeze et al.        | N/A     | N/A           |
| 2011/0087081 | 12/2010 | Kiani et al.        | N/A     | N/A           |
| 2011/0118561 | 12/2010 | Tari et al.         | N/A     | N/A           |
| 2011/0137297 | 12/2010 | Kiani et al.        | N/A     | N/A           |
| 2011/0172498 | 12/2010 | Olsen et al.        | N/A     | N/A           |
| 2011/0172743 | 12/2010 | Davis et al.        | N/A     | N/A           |
| 2011/0183650 | 12/2010 | McKee               | 455/567 | H04M<br>1/724 |
| 2011/0230733 | 12/2010 | Al-Ali              | N/A     | N/A           |
| 2012/0123231 | 12/2011 | O'Reilly            | N/A     | N/A           |
| 2012/0165629 | 12/2011 | Merritt et al.      | N/A     | N/A           |
| 2012/0209084 | 12/2011 | Olsen et al.        | N/A     | N/A           |
| 2012/0215559 | 12/2011 | Flanagan            | 705/3   | G16H<br>15/00 |
| 2012/0226117 | 12/2011 | Lamego et al.       | N/A     | N/A           |
| 2012/0238854 | 12/2011 | Blomquist et al.    | N/A     | N/A           |
| 2012/0277543 | 12/2011 | Homchowdhury et al. | N/A     | N/A           |
| 2012/0283524 | 12/2011 | Kiani et al.        | N/A     | N/A           |
| 2013/0023775 | 12/2012 | Lamego et al.       | N/A     | N/A           |
| 2013/0041591 | 12/2012 | Lamego              | N/A     | N/A           |
| 2013/0060147 | 12/2012 | Welch et al.        | N/A     | N/A           |
| 2013/0096405 | 12/2012 | Garfio              | N/A     | N/A           |
| 2013/0162424 | 12/2012 | Treacy              | N/A     | N/A           |
| 2013/0237770 | 12/2012 | Sullivan et al.     | N/A     | N/A           |
| 2013/0267791 | 12/2012 | Halperin et al.     | N/A     | N/A           |
| 2013/0296672 | 12/2012 | O'Neil et al.       | N/A     | N/A           |
| 2013/0345921 | 12/2012 | Al-Ali et al.       | N/A     | N/A           |
| 2014/0142963 | 12/2013 | Hill                | 705/2   | G16H<br>10/60 |
| 2014/0166076 | 12/2013 | Kiani et al.        | N/A     | N/A           |
| 2014/0180160 | 12/2013 | Brown et al.        | N/A     | N/A           |

|              |         |                  |     |     |
|--------------|---------|------------------|-----|-----|
| 2014/0187973 | 12/2013 | Brown et al.     | N/A | N/A |
| 2014/0275871 | 12/2013 | Lamego et al.    | N/A | N/A |
| 2014/0275872 | 12/2013 | Merritt et al.   | N/A | N/A |
| 2014/0316217 | 12/2013 | Purdon et al.    | N/A | N/A |
| 2014/0316218 | 12/2013 | Purdon et al.    | N/A | N/A |
| 2014/0323897 | 12/2013 | Brown et al.     | N/A | N/A |
| 2014/0323898 | 12/2013 | Purdon et al.    | N/A | N/A |
| 2015/0005600 | 12/2014 | Blank et al.     | N/A | N/A |
| 2015/0011907 | 12/2014 | Purdon et al.    | N/A | N/A |
| 2015/0073241 | 12/2014 | Lamego           | N/A | N/A |
| 2015/0080754 | 12/2014 | Purdon et al.    | N/A | N/A |
| 2015/0099950 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 2015/0106121 | 12/2014 | Muhsin et al.    | N/A | N/A |
| 2015/0186608 | 12/2014 | Fuller et al.    | N/A | N/A |
| 2015/0257654 | 12/2014 | Bennett-Guerrero | N/A | N/A |
| 2015/0310733 | 12/2014 | Gross            | N/A | N/A |
| 2016/0196388 | 12/2015 | Lamego           | N/A | N/A |
| 2016/0367173 | 12/2015 | Dalvi et al.     | N/A | N/A |
| 2017/0024748 | 12/2016 | Haider           | N/A | N/A |
| 2017/0042488 | 12/2016 | Muhsin           | N/A | N/A |
| 2017/0055896 | 12/2016 | Al-Ali           | N/A | N/A |
| 2017/0173632 | 12/2016 | Al-Ali           | N/A | N/A |
| 2017/0251974 | 12/2016 | Shreim et al.    | N/A | N/A |
| 2017/0311891 | 12/2016 | Kiani et al.     | N/A | N/A |
| 2018/0103874 | 12/2017 | Lee et al.       | N/A | N/A |
| 2018/0199871 | 12/2017 | Pauley et al.    | N/A | N/A |
| 2018/0242926 | 12/2017 | Muhsin et al.    | N/A | N/A |
| 2018/0247353 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 2018/0247712 | 12/2017 | Muhsin et al.    | N/A | N/A |
| 2018/0256087 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 2018/0296161 | 12/2017 | Shreim et al.    | N/A | N/A |
| 2018/0300919 | 12/2017 | Muhsin et al.    | N/A | N/A |
| 2018/0310822 | 12/2017 | Indorf et al.    | N/A | N/A |
| 2018/0310823 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 2018/0317826 | 12/2017 | Muhsin et al.    | N/A | N/A |
| 2019/0015023 | 12/2018 | Monfre           | N/A | N/A |
| 2019/0117070 | 12/2018 | Muhsin et al.    | N/A | N/A |
| 2019/0200941 | 12/2018 | Chandran et al.  | N/A | N/A |
| 2019/0239787 | 12/2018 | Pauley et al.    | N/A | N/A |
| 2019/0320906 | 12/2018 | Olsen            | N/A | N/A |
| 2019/0374139 | 12/2018 | Kiani et al.     | N/A | N/A |
| 2019/0374173 | 12/2018 | Kiani et al.     | N/A | N/A |
| 2019/0374713 | 12/2018 | Kiani et al.     | N/A | N/A |
| 2020/0021930 | 12/2019 | Iswanto et al.   | N/A | N/A |
| 2020/0060869 | 12/2019 | Telfort et al.   | N/A | N/A |
| 2020/0111552 | 12/2019 | Ahmed            | N/A | N/A |
| 2020/0113435 | 12/2019 | Muhsin           | N/A | N/A |
| 2020/0113488 | 12/2019 | Al-Ali et al.    | N/A | N/A |
| 2020/0113496 | 12/2019 | Scruggs et al.   | N/A | N/A |
| 2020/0113497 | 12/2019 | Triman et al.    | N/A | N/A |

|              |         |                      |     |     |
|--------------|---------|----------------------|-----|-----|
| 2020/0113520 | 12/2019 | Abdul-Hafiz et al.   | N/A | N/A |
| 2020/0138288 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0138368 | 12/2019 | Kiani et al.         | N/A | N/A |
| 2020/0163597 | 12/2019 | Dalvi et al.         | N/A | N/A |
| 2020/0196877 | 12/2019 | Vo et al.            | N/A | N/A |
| 2020/0253474 | 12/2019 | Muhsin et al.        | N/A | N/A |
| 2020/0253544 | 12/2019 | Belur Nagaraj et al. | N/A | N/A |
| 2020/0275841 | 12/2019 | Telfort et al.       | N/A | N/A |
| 2020/0288983 | 12/2019 | Telfort et al.       | N/A | N/A |
| 2020/0321793 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0329983 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0329984 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0329993 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0330037 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2021/0022628 | 12/2020 | Telfort et al.       | N/A | N/A |
| 2021/0104173 | 12/2020 | Pauley et al.        | N/A | N/A |
| 2021/0113121 | 12/2020 | Diab et al.          | N/A | N/A |
| 2021/0117525 | 12/2020 | Kiani et al.         | N/A | N/A |
| 2021/0118581 | 12/2020 | Kiani et al.         | N/A | N/A |
| 2021/0121582 | 12/2020 | Krishnamani et al.   | N/A | N/A |
| 2021/0161465 | 12/2020 | Barker et al.        | N/A | N/A |
| 2021/0236729 | 12/2020 | Kiani et al.         | N/A | N/A |
| 2021/0256267 | 12/2020 | Ranasinghe et al.    | N/A | N/A |
| 2021/0256835 | 12/2020 | Ranasinghe et al.    | N/A | N/A |
| 2021/0275101 | 12/2020 | Vo et al.            | N/A | N/A |
| 2021/0290060 | 12/2020 | Ahmed                | N/A | N/A |
| 2021/0290072 | 12/2020 | Forrest              | N/A | N/A |
| 2021/0290080 | 12/2020 | Ahmed                | N/A | N/A |
| 2021/0290120 | 12/2020 | Al-Ali               | N/A | N/A |
| 2021/0290177 | 12/2020 | Novak, Jr.           | N/A | N/A |
| 2021/0290184 | 12/2020 | Ahmed                | N/A | N/A |
| 2021/0296008 | 12/2020 | Novak, Jr.           | N/A | N/A |
| 2021/0330228 | 12/2020 | Olsen et al.         | N/A | N/A |
| 2021/0386382 | 12/2020 | Olsen et al.         | N/A | N/A |
| 2021/0402110 | 12/2020 | Pauley et al.        | N/A | N/A |
| 2022/0026355 | 12/2021 | Normand et al.       | N/A | N/A |
| 2022/0039707 | 12/2021 | Sharma et al.        | N/A | N/A |
| 2022/0053892 | 12/2021 | Al-Ali et al.        | N/A | N/A |
| 2022/0071562 | 12/2021 | Kiani                | N/A | N/A |
| 2022/0096603 | 12/2021 | Kiani et al.         | N/A | N/A |
| 2022/0151521 | 12/2021 | Krishnamani et al.   | N/A | N/A |
| 2022/0218244 | 12/2021 | Kiani et al.         | N/A | N/A |
| 2022/0287574 | 12/2021 | Telfort et al.       | N/A | N/A |
| 2022/0296161 | 12/2021 | Al-Ali et al.        | N/A | N/A |
| 2022/0361819 | 12/2021 | Al-Ali et al.        | N/A | N/A |
| 2022/0379059 | 12/2021 | Yu et al.            | N/A | N/A |
| 2022/0392610 | 12/2021 | Kiani et al.         | N/A | N/A |
| 2023/0028745 | 12/2022 | Ai-Ali               | N/A | N/A |
| 2023/0038389 | 12/2022 | Vo                   | N/A | N/A |
| 2023/0045647 | 12/2022 | Vo                   | N/A | N/A |

|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2023/0058052 | 12/2022 | Al-Ali             | N/A | N/A |
| 2023/0058342 | 12/2022 | Kiani              | N/A | N/A |
| 2023/0069789 | 12/2022 | Koo et al.         | N/A | N/A |
| 2023/0087671 | 12/2022 | Telfort et al.     | N/A | N/A |
| 2023/0110152 | 12/2022 | Forrest et al.     | N/A | N/A |
| 2023/0111198 | 12/2022 | Yu et al.          | N/A | N/A |
| 2023/0115397 | 12/2022 | Vo et al.          | N/A | N/A |
| 2023/0116371 | 12/2022 | Mills et al.       | N/A | N/A |
| 2023/0135297 | 12/2022 | Kiani et al.       | N/A | N/A |
| 2023/0138098 | 12/2022 | Telfort et al.     | N/A | N/A |
| 2023/0145155 | 12/2022 | Krishnamani et al. | N/A | N/A |
| 2023/0147750 | 12/2022 | Barker et al.      | N/A | N/A |
| 2023/0210417 | 12/2022 | Al-Ali et al.      | N/A | N/A |
| 2023/0222805 | 12/2022 | Muhsin et al.      | N/A | N/A |
| 2023/0222887 | 12/2022 | Muhsin et al.      | N/A | N/A |
| 2023/0226331 | 12/2022 | Kiani et al.       | N/A | N/A |
| 2023/0284916 | 12/2022 | Telfort            | N/A | N/A |
| 2023/0284943 | 12/2022 | Scruggs et al.     | N/A | N/A |
| 2023/0301562 | 12/2022 | Scruggs et al.     | N/A | N/A |
| 2023/0346993 | 12/2022 | Kiani et al.       | N/A | N/A |
| 2023/0368221 | 12/2022 | Haider             | N/A | N/A |
| 2023/0371893 | 12/2022 | Al-Ali et al.      | N/A | N/A |
| 2023/0389837 | 12/2022 | Krishnamani et al. | N/A | N/A |
| 2024/0016418 | 12/2023 | Devadoss et al.    | N/A | N/A |
| 2024/0016419 | 12/2023 | Devadoss et al.    | N/A | N/A |
| 2024/0047061 | 12/2023 | Al-Ali et al.      | N/A | N/A |
| 2024/0049310 | 12/2023 | Al-Ali et al.      | N/A | N/A |
| 2024/0049986 | 12/2023 | Al-Ali et al.      | N/A | N/A |
| 2024/0081656 | 12/2023 | DeJong et al.      | N/A | N/A |
| 2024/0122486 | 12/2023 | Kiani              | N/A | N/A |
| 2024/0180456 | 12/2023 | Al-Ali             | N/A | N/A |
| 2024/0188872 | 12/2023 | Al-Ali et al.      | N/A | N/A |
| 2024/0245855 | 12/2023 | Vo et al.          | N/A | N/A |
| 2024/0260894 | 12/2023 | Olsen              | N/A | N/A |
| 2024/0267698 | 12/2023 | Telfort et al.     | N/A | N/A |
| 2024/0277233 | 12/2023 | Al-Ali             | N/A | N/A |
| 2024/0277280 | 12/2023 | Al-Ali             | N/A | N/A |
| 2024/0298920 | 12/2023 | Fernkbist et al.   | N/A | N/A |
| 2024/0306985 | 12/2023 | Vo et al.          | N/A | N/A |
| 2024/0324953 | 12/2023 | Telfort            | N/A | N/A |
| 2024/0380246 | 12/2023 | Moran              | N/A | N/A |
| 2024/0380247 | 12/2023 | Moran              | N/A | N/A |
| 2025/0000458 | 12/2024 | Abdul-Hafiz et al. | N/A | N/A |

#### FOREIGN PATENT DOCUMENTS

| Patent No.   | Application Date | Country | CPC |
|--------------|------------------|---------|-----|
| WO 98/038909 | 12/1997          | WO      | N/A |
| WO 02/051354 | 12/2001          | WO      | N/A |
| WO 03/094090 | 12/2002          | WO      | N/A |

|                |         |    |     |
|----------------|---------|----|-----|
| WO 2008/116295 | 12/2007 | WO | N/A |
| WO 2013/056180 | 12/2012 | WO | N/A |
| WO 2013/076481 | 12/2012 | WO | N/A |

## OTHER PUBLICATIONS

US 2022/0192529 A1, 06/2022, Al-Ali et al. (withdrawn) cited by applicant

US 2024/0016391 A1, 01/2024, Lapotko et al. (withdrawn) cited by applicant

Cvach, Maria, "Monitor Alarm Fatigue: An Integrative Review", Biomedical Instrumentation & Technology, July/Aug. 2012, pp. 268-277. cited by applicant

Lorincz et al., "Sensor Networks for Emergency Response: Challenges and Opportunities", Pervasive Computing, Oct.-Dec. 2004, pp. 16-23. cited by applicant

Lukowicz et al., "Amon: A Wearable Medical Computer for High Risk Patients", Proceedings of the 6th International Symposium on Wearable Computers (ISWC'02), pp. 2. cited by applicant

*Primary Examiner:* Tun; Nay

*Attorney, Agent or Firm:* Knobbe, Martens, Olson & Bear, LLP

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/028,822, filed on Sep. 22, 2020, which is a continuation of U.S. patent application Ser. No. 16/735,476, filed on Jan. 6, 2020, now issued as U.S. Pat. No. 10,813,580 on Oct. 27, 2020, entitled ALARM FATIGUE MANAGEMENT SYSTEMS AND METHODS, which is a continuation of U.S. patent application Ser. No. 16/155,006, filed on Oct. 9, 2018, now issued as U.S. Pat. No. 10,524,712 on Jan. 7, 2020, and entitled ALARM FATIGUE MANAGEMENT SYSTEMS AND METHODS, which is a continuation of U.S. patent application Ser. No. 14/738,658, filed on Jun. 12, 2015, now issued as U.S. Pat. No. 10,123,729 on Nov. 13, 2018, and entitled ALARM FATIGUE MANAGEMENT SYSTEMS AND METHODS which relates to and claims the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 62/011,643, filed on Jun. 13, 2014 and entitled ALARM FATIGUE MANAGEMENT SYSTEMS AND METHODS, the disclosures of each of which are hereby incorporated by reference in their entirety.

## TECHNICAL FIELD

(1) This disclosure relates in general to the field of healthcare systems and, more particularly, to systems and methods related to alarm fatigue management.

## BACKGROUND

(2) The background description includes information that may be useful in understanding the present disclosure. It is not an admission that any of the information provided herein is prior art or relevant to the disclosure, or that any publication specifically or implicitly referenced is prior art.

(3) Alarm fatigue is desensitization to alarms brought on by overexposure to excessive alarms, which can result in reduced response times or even complete failure to respond to critical issues that are raised by the alarms in the first place. Alarm fatigue is increasingly a serious problem in a variety of different industries and professions. In particular, medical professionals can experience alarm fatigue so severe that alarms indicating life threatening conditions are at times overlooked, resulting in numerous deaths each year. Various systems tailored towards modifying delivery of alarms exist in the healthcare marketplace. One such system creates alert signals based on information or data from medical systems. The alert signals can take the form of music that is

generated using the information or data from the medical system, creating a wide variety of signals that simultaneously pass information to an intended recipient. Another system reduces false alarms in a certain predetermined region around a medical device. A medical professional has a portable transmitter/monitor, and when an alert condition exists, the system will check the physical proximity of the transmitter/monitor. In the event the transmitter/monitor is within the predetermined region defined as a false alarm region, the alert is concealed.

(4) In yet another example, portable alert devices deliver alerts to an intended recipient. The portable alert devices perform a scan to gather relevant information about the device's surroundings prior to issuing an alert. In doing so, they can alter the mode of an alert depending on the environment that the portable devices and the intended recipient are in. In yet another example device, when a user is supposed to pump insulin, the device provides an alarm. To reduce alarm fatigue, the device includes a randomization module that can generate random alarms. Randomness of alarms is determined by historical stability of the user's blood glucose level. Thus, a user with a more stable blood glucose level will be rewarded with fewer alerts to check their blood glucose level. Another system creates "super-alarms."

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) To provide a more complete understanding of the present disclosure and features and advantages thereof, reference is made to the following description, taken in conjunction with the accompanying figures, wherein like reference numerals represent like parts, in which:
- (2) FIG. 1 is a simplified block diagram illustrating a system for alarm fatigue management according to an example embodiment;
- (3) FIG. 2 is a simplified block diagram illustrating example details of the system according to an embodiment;
- (4) FIG. 3 is a simplified block diagram illustrating other example details of the system according to an embodiment;
- (5) FIG. 4 is a simplified block diagram illustrating yet other example details of the system according to an embodiment;
- (6) FIG. 5 is a simplified block diagram illustrating yet other example details of the system according to an embodiment;
- (7) FIG. 6 is a simplified block diagram illustrating yet other example details of the system according to an embodiment;
- (8) FIG. 7 is a simplified flow diagram illustrating example operations that may be associated with an embodiment of the system;
- (9) FIG. 8 is a simplified block diagram illustrating yet other example details of the system according to an embodiment;
- (10) FIG. 9 is a simplified flow diagram illustrating other example operations that may be associated with an embodiment of the system; and
- (11) FIG. 10 is a simplified diagram illustrating example details according to an embodiment of the system.

### DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

#### Overview

(12) An apparatus for modifying alarms at a medical device for alarm fatigue management is provided and includes an alarm monitor, an alarm filter, an alarm modifier, a memory element for storing data, and a processor that executes instructions associated with the data, wherein the processor and the memory element cooperate such that the apparatus is configured for receiving an alarm condition from an alarm management engine, the alarm condition based on an alarm fatigue

level of a user of the medical device, the alarm fatigue level based on at least a user fatigue model, a medical device model and a patient condition, receiving an alarm from the medical device, modifying the alarm according to the alarm condition, the alarm condition being configured to increase a likelihood of the user responding to the modified alarm, and generating an alarm indicator based on the modification.

#### EXAMPLE EMBODIMENTS

(13) Turning to FIG. 1, FIG. 1 is a simplified block diagram illustrating a system **10** according to an example embodiment. System **10** comprises a network **11** (generally indicated by an arrow) connecting an alarm management engine **12** with one or more user devices **14** and medical devices **16**. Alarm management engine **12** interfaces with a user fatigue model database **18**, a medical device model database **20**, and a patient conditions database **22**. User fatigue model database **18** can include a plurality of user fatigue model **24** corresponding to different users or stakeholders of system **10**. Each user fatigue model **24** can include one or more user fatigue model attribute **26**.

(14) User fatigue model attribute **26** can comprise factors (e.g., characteristics) that can affect a user's fatigue level. User fatigue model attribute **26** can include many different types of information, including information conveyed via status data **27** from user device **14** and information entered into user fatigue model database **18** by other means, based on details about a particular user. In various embodiments, status data **27** can modify user fatigue model attribute **26** of user fatigue model **24**. For example, user fatigue model attribute **26** comprises a number of hours worked by the user; status data **27** can indicate that the user worked for 8 hours. In another example, user fatigue model attribute **26** comprises a shift information for users on a particular floor of a hospital; status data **27** can indicate that the shift is a night shift.

(15) In a general sense, status data **27** comprises data corresponding to various measurable quantities related to a particular user. Status data **27** can include physiological information (e.g., stress level (as measured by various measurable parameters such as blood pressure, heart rate, etc.), neurological activity, brain electrical activity, hormone levels, body temperature, breathing rate, blood sugar level, age, change of position, sleep activity (e.g., amount of sleep over previous 24 hours and time since last sleep), etc.), alarm response time information, user shift data (e.g., if user is not on active duty, user may ignore alarm), environmental alarm data (e.g., number of alarms received at user device), and various other parameters that can indicate a likelihood of the corresponding user responding to, or ignoring, an alarm signal. In some embodiments, status data **27** can be communicated with an associated timestamp corresponding to a time of data collection/measurement. In some embodiments, one or more correlations between a physiological reaction and an alarm signal may be discerned.

(16) User shift data can include various details describing the user's shift, including the user's work schedule (e.g., weekly schedule, monthly schedule, yearly schedule, number of shifts worked without a day off, length of a shift, total time elapsed since the shift began, and/or number of shifts worked since the last vacation). For example, user shift data can indicate when a user works a late night shift followed by an early morning shift the next day. Status data **27** can also include an amount of user-directed alarms (e.g., the number of alarms the user has had to respond to), alarm response time data (e.g., the user's time to respond), alarm severity for at least one user-directed alarm, a number of repeated alarms (e.g., the number of alarms that have had to be repeated due to a delayed response or lack of a response), a number of missed alarms, a number of simultaneous alarms, a number of false alarms, etc.

(17) Environment alarm data can include the number of alarms the user has been exposed to (e.g., both alarms directed to the user and alarms directed to other users), the loudness of an alarm the user has been exposed to, the frequency of alarms the user has been exposed to during a relevant time period (e.g., during a shift, during a day, during a week, during two weeks, during a month, during a year, since time of most recent vacation, and since time of most recent day off), and at least one alarm location (e.g., a location where the user heard an alarm).

(18) In some embodiments, user fatigue model attribute **26** can be changed (e.g., updated) periodically or continuously, depending on the sampling frequency in an embodiment. Additionally, user fatigue model attribute **26** can contain historical data (e.g., longitudinal study data). Some examples of user fatigue model attribute **24** include personality factors and physiological factors. Personality factors are factors that might indicate a person is more or less prone to alarm fatigue (e.g., irritability, psychological disorders, and personality type). Physiological factors can include factors that pertain to a certain user's physical abilities (e.g., deafness, blindness, color blindness, sensitivity to particular light and/or sound patters, and varying degrees of disabilities related to these conditions and others).

(19) Medical device model database **20** can include one or more medical device model **28**. Medical device model **28** can comprise a data construct (e.g., an algorithmic model, a mathematical model, a digital model, an alphanumeric identifier, etc.) of medical device **16**, a representation of an instrument reading (e.g., blood pressure measurement), or other suitable function. Note that the term “data construct” as used herein encompasses scalars, arrays, subarrays, matrices, vectors, and other data representations that are allocated in a region of memory in a memory element. The data constructs as used herein represent analog information (such as physical phenomena) with binary digits (or other computer interpretable representation) that are interpreted as integers, real numbers, characters, or other data types comprising a finite number of discrete symbols that represent essentially infinite variation of analog information. In other words, the data construct represents a physical entity in symbolic form, typically using symbols from a relatively small number of discrete enumerable symbols associated with a context and other properties of the physical entity being represented.

(20) In some embodiments, medical device model **28** can comprise a data container for numerical values corresponding to device readings (e.g., measurements). Each medical device model **28** can include one or more alarm condition parameter **30**. For example, one of medical device model **28** can comprise a model of a blood pressure monitor; alarm condition parameter **30** for such model may correspond to a blood pressure reading that indicates an alarm if the value of alarm condition parameter **30** exceeds a predetermined threshold or other criteria. In some embodiments, the threshold may be included in corresponding medical device model **28**.

(21) In another example, alarm condition parameter **30** comprises a threshold value for a physiological measurement of a patient measured by medical device **16**. In yet another example, alarm condition parameter **30** comprises a maximum frequency at which medical device **16** generates an alarm. In yet another example, alarm condition parameter **30** comprises a frequency at which medical device **16** measures the patient. In yet another example, alarm condition parameter **30** comprises measurement sensitivity corresponding to medical device **16**'s ability to measure the patient. In various embodiments, each medical device **16** may be represented by corresponding medical device model **28** in medical device model database **20**.

(22) Patient conditions database **22** can include one or more patient condition **32**. By way of examples, and not as limitations, patient condition **32** may include a data construct of an aggregate (e.g., collection, average, weighted average, composite, etc.) of health conditions, population characteristics (e.g., lack of dexterity or memory with older users; high mental workload for anesthesiologists in operating rooms; etc.), diseases, symptoms (e.g., subjective patient symptoms and observable symptoms such as temperature, analyte data, etc.), medications and various medical status. Examples of patient condition **32** include physical or mental health parameter values, such as blood glucose in diabetes, respiratory flow in asthma, blood pressure in hypertension, cholesterol in cardiovascular disease, weight in eating disorders, T-cell or viral count in HIV, and frequency or timing of episodes in mental health disorders.

(23) In a general sense, patient condition **32** is indicative of a context of the physiological measurement by medical device **16** (e.g., context under which the alarms are generated at medical device **16**). For example, alarms generated at medical device **16** may indicate a hypertension status

of a patient when the alarms are directed towards, or result as a consequence of, high blood pressure of the patient. In another example, alarms generated at medical device **16** may indicate a serious condition of the patient when the alarms are directed towards, or result as a consequence of, various disparate physiological measurements of the patient.

(24) In another example, patient condition **32** can refer to a medical status of the patient (e.g., “undetermined” may correspond to a status of a patient awaiting physician and assessment; “good” may indicate that vital signs (e.g., pulse rate, oxygen levels, blood pressure, etc.) are stable and within normal limits, with the patient being conscious and comfortable; “fair” may indicate that vital signs are stable and within normal limits with the patient being conscious, but uncomfortable; “serious” may indicate that vital signs are unstable and not within normal limits, with the patient being acutely ill; “critical” may indicate that vital signs are unstable and not within normal limits with the patient being unconscious; “deceased” may indicate that the patient is dead). In another example, patient condition **32** of “hypertension” may indicate that the patient suffers from hypertension, with expected high blood pressure readings; patient condition **32** of “diabetes” may indicate that the patient suffers from diabetes with higher than normal blood sugar readings without medication, etc. In some embodiments, patient condition **32** may correspond to a mathematical model (e.g., polynomial function, alphanumeric value, matrix, etc.) of a corresponding physical or mental health condition.

(25) In some embodiments, alarm management engine **12** receives, in addition to user device status data **27**, one or more alarm indicator **34** from one or more medical device **16**. Alarm indicator **34** can indicate modifications to alarms generated at medical device **16**, and can serve as feedback regarding the modifications (e.g., whether the modifications are effective). In some embodiments, alarm indicator **34** can also include signals (e.g., auditory, tactile, optical, vibratory, electrical, wireless, and/or other signals) that are indicative of, or associated with alarms configured on medical device **16**. Alarm management engine **12** may include, in addition to a processor **36** and a memory element **38**, an alarm fatigue calculator **40** and an alarm condition calculator **42**. Alarm management engine **12** may be coupled to (and communicate with) adapter **46** associated with respective medical device **16**.

(26) During operation, alarm management engine **12** may receive status data **27** from user device **14**. Status data **27** can be collected substantially continuously in some embodiments, and periodically (e.g., every 1, 5, 10, 15, 20, 30, 45 second and/or every 1, 5, 10, 15, 30, 45 minutes) in other embodiments. In some embodiments, status data **27** may be collected in predetermined time intervals or ranges (e.g., every 0-5 seconds, 5-10 seconds, 10-15 seconds, 15-20 seconds, 20-30 seconds, 30-45 seconds, 45-60 seconds and/or every 0-5, 5-10, 10-15, 15-30, or 30-45 minutes). Additionally, status data **27** may be collected on a “push” basis (e.g., in response to certain triggers such as alerts and responses to alerts) with user device **14** pushing status data **27** to alarm management engine **12**. Alarm fatigue calculator **40** may calculate alarm fatigue levels associated with respective users of user device **14** based on one or more user fatigue model **24** in user fatigue model database **18**.

(27) Alarm fatigue calculator **40** may feed the calculated alarm fatigue level to alarm condition calculator **42**. Alarm condition calculator **42** may calculate alarm condition **44** for medical device **16** based on information from status data **27**, alarm indicator **34**, calculated alarm fatigue level, medical device model **28**, and patient condition **32**. In various embodiments, alarm condition **44** comprises instructions for alarms generated by medical device **16**. The instructions can specify alarm attributes (e.g., alarm frequency, alarm threshold, alarm distribution mode (such as visual, auditory, sensory, tactile, etc.), alarm intensity (e.g., for a given distribution mode, how intensely the alarm is presented), alarm duration, alarm severity, etc.), alarm output interface (e.g., on user device **14**, on a centralized alarm system, or medical device **16**, etc.), measurement sensitivity, and any other parameter that affects likelihood of increased response to the alarms by one or more users whose alarm fatigue levels are considered in generating alarm condition **44**.

(28) Alarm condition **44** may be communicated to an adapter **46** located at applicable medical device **16**. Adapter **46** may monitor and filter alarms from medical device **16** based on alarm condition **44**. In some embodiments, medical device **16** and/or adapter **46** may generate and communicate alarm indicator **34** to alarm management engine **12**. Alarm indicator **34** can provide feedback about alarm condition **44**, which can be used in machine learning algorithms.

(29) In some embodiments, status data **27** can indicate that an alarm has been falsely generated. Alarm management engine **12** can use the false alarm indication information to learn conditions that lead to false alarms generation. To achieve condition learning, embodiments of alarm management engine **12** can implement various machine learning algorithms. For example, some embodiments implement supervised learning techniques, such as Averaged One-Dependence Estimators (AODE), artificial neural network, Bayesian statistics, case-based reasoning, decision trees, inductive logic programming, Gaussian process regression, gene expression programming, group method of data handling, learning automata, learning vector quantization, logistic model tree, minimum message length, lazy learning, instance-based learning, probably approximately correct learning, ripple down rules, symbolic machine learning, sub-symbolic machine learning algorithms, support vector machines, random forests, ensembles of classifiers, bootstrap aggregating (bagging), boosting (meta-algorithm), ordinal classification, regression analysis, information fuzzy networks (IFN), and conditional random field. Suitable sources for machine learning systems include those available at [www.scikit-learn.org](http://www.scikit-learn.org)), which offers numerous machine learning, data mining, and data analysis tools based on the Python computer language.

(30) In other embodiments, alarm management engine **12** can also (e.g., alternatively, or additionally) implement unsupervised learning techniques, such as artificial neural network, data clustering, expectation-maximization algorithm, self-organizing map, radial basis function network, vector quantization, generative topographic map, information bottleneck method, association rule learning, Apriori algorithm, Eclat algorithm, Frequent Pattern (FP)-growth algorithm, hierarchical clustering, single-linkage clustering, conceptual clustering, partitional clustering, k-means algorithm, fuzzy clustering, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Ordering Points To Identify the Clustering Structure (OPTICS) algorithm, outlier detection, local outlier factor, reinforcement learning, temporal difference learning, q-learning, learning automata, Monte Carlo method, Sarsa, deep learning, deep belief networks, deep Boltzmann machines, deep convolutional neural networks, and deep recurrent neural networks.

(31) In embodiments, one or more machine learning algorithms can be used to create a smart alarm management system that can develop rules for alarm generation. For example, alarm indicator **34** and/or status data **27** indicative of a false alarm can be aggregated over a period of time allowing a machine learning algorithm executing in alarm management engine **12** to learn when an alarm is more or less likely to be a false alarm. Over time, the algorithm can additionally learn when to prevent generation of an alarm based on historical information and trends learnt from alarm indicator **34** and status data **27**. In another example, a smart alarm system could be used to determine whether an alarm should be generated based on data collected from plurality of medical device **16**.

(32) In some embodiments, alarm management engine **12** can implement one or more machine learning algorithms to develop prediction models for alarms. For example, alarm management engine **12** can provide a projected number of false alarms for a particular shift and it can take the calculated projected number of false alarms into account by creating rules for generating alarms throughout that shift.

(33) In some embodiments, users can manually input status data to the system. For example, a user can report dreams or nightmares. A user can also report diet, exercise, and different types of ailments (e.g., if a user gets the flu, the user could report that to the system). In various embodiments, alarm management engine **12** can receive a variety of user-created status data inputs in status data **27** that may be used to create better (e.g., more productive, useful, etc.) user fatigue

models **24**.

(34) In some embodiments, user fatigue model **24** is updated based on status data **27**. Multiple user fatigue models **24** can be updated simultaneously or in sequence as status data **27** is received; in some embodiments, alarm management engine **12** may be limited in the updating by the number of users registered thereto. In some embodiments, incoming status data **27** can overwrite previously stored status data within user fatigue model **24**; in other embodiments, incoming status data **27** can be stored along with previously stored status data to create a historical set of status data.

(35) In various embodiments, the alarm fatigue level may be calculated based on previously updated user fatigue model **24**. Alarm fatigue level can have a baseline and a current value. For example, one or more users might have a slowly increasing alarm fatigue level week over week. However, daily factors impacting the user can alter (e.g., add to or subtract from) the base line. For example, a base line can increase by 2% week over week. In another example, a user is alert with a reduced fatigue relative to the baseline in the morning, but after a 12 hour shift the user's fatigue level is heightened relative to that user's baseline. A baseline might be reset after vacation, for example.

(36) A “default” user fatigue model **24** for a particular user can comprise the baseline in some embodiments. The baseline can be a historically determined “normal” fatigue level for the particular user based on samples collected during a day, a week, a particular shift, and/or a particular length of shift, all of which serve to determine an “un-fatigued” state. For example, as the day progresses, user fatigue model **24** can be updated based on various status data **27** received and the user's alarm fatigue level can subsequently be determined. As user fatigue model **24** of the corresponding user is updated, the alarm given to the user can be adapted accordingly. Consider a scenario where a user, say a nurse or doctor, has numerous long shifts. The disclosed system might discover that the user becomes fatigued with respect to alarms after more than 10 hours of shift on a consistent basis. The system discovers this condition by measuring the user's response time to alarms where the response time begins to increase measurably after 10 hours. Even though the user is responding within an acceptable period, the measured increase in response time can be considered as a leading indicator for fatigue. Discovery of this leading indicator can then be considered part of the specific user's baseline fatigue model in future settings. Further, such a measured indicator can also be incorporated into the fatigue models of other stakeholders having similar characteristics of the user; say all nurses or all doctors for example, if the measured indicator is validated for the class of users.

(37) Using status data **27**, some embodiments alarm management engine **12** can adapt and learn how quickly a particular user's alarm fatigue level increases over time given normal operating (e.g., working) conditions. Other factors that can be learned over time include how the user's alarm fatigue level changes after some number of days off, after a weekend, and/or after a night of sleep (e.g., where the user gets only 6 hours of sleep compared to when the user gets 10 hours of sleep).

(38) In some embodiments, the alarm fatigue level can be categorized into different tiers, such as no alarm fatigue, low-level alarm fatigue, mid-level alarm fatigue, and high-level alarm fatigue. In other embodiments, the alarm fatigue level can be classified along a continuum. When alarm fatigue is classified along a continuum, alarms generated may be modified based on user fatigue model attributes **26** seen as contributing more to alarm fatigue levels. For example, if the particular user experiences alarm fatigue caused mainly by a high number of environment alarms, alarm management engine **12** can provide the user with specially tailored alarms. In another example, if the user has experienced a high number of alarms that incorporate particular sounds, the system can provide an alarm that includes a different set of sounds that may also incorporate tactile feedback and/or visual feedback. User fatigue attributes **26** can be weighted according to importance or strength of correlations in some embodiments. In some embodiments, alarm fatigue levels can be characterized on different scales, such as personalized fatigue level (e.g., specific to a particular user), a group level (e.g., specific to users working in a particular group), a “floor” level (e.g.,

specific to users working in a particular floor level of a building), a “shift” level (e.g., specific to users working in a particular shift), an institution level (e.g., specific to users working in a particular institution), etc. In some embodiments, alarm fatigue levels can have many dimensions beyond a tiered system. Such an approach can be considered advantageous, for example, because it allows the system to deliver alarms to the user where the alarm not only addresses the user's fatigue, but also provides an indication to the user that they are, in fact, suffering from fatigue based on the modality or nature of the alarm.

(39) In a general sense, fatigue data can be classified into various types, examples include emotional fatigue, physical fatigue, or mental acuity fatigue. Each fatigue-type can be inferred based on status data **27**. For example, mental acuity fatigue may be inferred by determining how long it takes the user to make a decision; physical fatigue may be inferred based on how long it physically takes the user to respond to an alarm or other activity (e.g., where response time can optionally be normalized based on distance traveled to respond to an alarm); etc. Each type of fatigue may correspond to a unique signature or template generated based on status data **27**.

(40) In various embodiments, alarm indicator **34** comprises an indication of a medical alert requiring a response from the user. In some embodiments, alarm indicator **34** may comprise inputs from medical device **16** that collect physiological information from the patient. When, for example, a medical sensor indicates a heart rate has flat lined, an alert condition is met. Alarm management engine **12** receives an indication of the alert condition through alarm indicator **34**.

(41) According to some embodiments, alarm management engine **12** may be capable of synthesizing and/or interpreting medical sensor data directly, and then determining whether an indication of a medical alert exists. In embodiments where the alarm management engine **12** is used with existing medical device **15**, alarm management engine **12** may learn and/or adapt so that alarm condition **44** can be standardized (e.g., made uniform) and managed regardless of the legacy and/or default alarm conditions used by medical device **16**. In some embodiments, learning models can be tied to patient condition **32**, care setting, device type, device manufacturer, and other such parameters.

(42) In various embodiments, alarm management engine **12** facilitates generating an alarm directed to a specific user taking into account the user's alarm fatigue level. In some embodiments, alarm management engine **12** additionally takes into account the alarm fatigue level of another user or users. In some embodiments, an alarm generation instruction is triggered by an indication of a medical alert from an outside source; thereafter, the alarm generation instruction is modified by one or more alarm fatigue levels; the modified alarm generation instruction generates the alarm. In various embodiments, alarm management engine **12** can take into account alarm fatigue levels of a single user; in other embodiments, alarm management engine **12** can take into account the alarm fatigue of a plurality of users. In various embodiments, alarm management engine **12** can also take into account alarm fatigue levels of non-users (e.g., persons outside the system may develop alarm fatigue, which can minimize the impact of the alarm based on the number of non-users potentially affected).

(43) According to some embodiments, alarm management engine **12** can take into account instructions that specific users should receive only particular alerts (e.g., some recipients can only interpret sound alerts generated in a particular frequency range). In various embodiments, system **10** can be configured to comply with various regulations (e.g., governmental, administrative, and private) with which alarm systems must comply.

(44) In some embodiments, when alarm fatigue level indicates no alarm fatigue, alarm condition **44** may permit various alarms to be generated (e.g., issued, created, sounded, etc.), for example, alarms based on sounds, vibrations, and/or a light, where the alarm corresponds to a “default” state. When alarm fatigue level indicates low-level alarm fatigue, alarm condition **44** may indicate a different alarm, for example, a specific combination of sounds, vibrations, and/or lights corresponding to the low-level alarm fatigue. Likewise, mid- and high-level alarm fatigue levels

may correspond to respective alarm types, such as unique sounds, lights, etc. tailored to alarm fatigue levels of relevant users.

(45) In some embodiments, the generated alarm may be distributed (e.g., disseminated, communicated, transmitted, sounded, etc.) based on alarm condition **44**. Distribution can be carried out in a number of different ways. For example, each individual user may have a personal alarm device (e.g., watch) that is capable of conveying an alarm perceptible to the intended recipient. In some embodiments, the distributed alarm is directed to relevant users to the exclusion of others.

(46) In an example scenario using embodiments of system **10**, consider a hypothetical intensive care unit (ICU) of a hospital, which is manned by two nurses, Alice and Bob. At 8:00 AM, nurse Alice logs into user device **14** (e.g., computer) at a central hub in the ICU to begin her day's shift. Status data **27** logs the time of entry and communicates it with Nurse Alice's credentials to alarm management engine **12**. Alarm fatigue calculator **40** calculates nurse Alice's user fatigue at level I (e.g., lowest level) based on user fatigue model **24** and user fatigue model attribute **26** determined from status data **27**.

(47) Assume that at 8:15 AM, a patient is brought into the ICU in serious condition. Nurse Alice registers the patient at the central hub and status data **27** indicative of the patient's condition is manually entered into user device **14** and received at alarm management engine **12**. Nurse Bob logs into an electronic chart at the patient's bedside and enters information about medications being provided to the patient into the electronic chart. Information entered by nurse Bob along with nurse Bob's credentials are sent to alarm management engine **12**. Assume that Nurse Bob is set to end his shift at 9:00 AM after 8 hours of continuous work at the ICU. Alarm fatigue calculator **40** calculates nurse Bob's user fatigue at level V (e.g., highest level) at the time of last receipt of status data **27** based on user fatigue model **24** and user fatigue model attribute **26** determined from status data **27**.

(48) Assume that medical device **16** substantially continuously measures the patient's blood pressure. A blood pressure reading at 8:16 AM indicates 130/90 mm Hg; at 8:20 AM indicates 120/80 mm Hg; and at 8:24 AM indicates 115/65 mmHg. Medical device model **28** corresponding to medical device **16** may specify that any blood pressure measurement over 120/80 mm Hg should generate an alarm. Patient condition **32** may specify that blood pressure for patients with a "serious" designation typically can vary beyond the normal range, but the rate of change is to be monitored more closely. Further in view of Bob's inferred high user fatigue level, generation of alarms may be tempered to only the more significant ones.

(49) Alarm condition calculator **42** generates alarm condition **44** based on information from medical device database **20**, patient conditions database **22** and user fatigue calculated by alarm fatigue calculator **40**, and communicates alarm condition **44** to an adapter **46** located at medical device **16**. Adapter **46** monitors and filters alarms from medical device **16**. In the example, alarms may not be generated at medical device **16** for blood pressure readings at 8:16 AM and 8:20 AM; however, alarms may be generated at 8:24 AM in view of the rate of change of blood pressure. In some embodiments, alarms may be generated at 8:16 AM and communicated to nurse Alice's work station, as nurse Alice has a lower inferred fatigue level than nurse Bob, but not displayed visibly or audibly at nurse Bob's work station, whereas alarms generated at 8:24 AM may be communicated to both nurse Alice and nurse Bob. Medical device **16** and/or adapter **46** may generate alarm indicator **34** indicating that alarms have been generated and/or communicated (e.g., based on alarm condition **44**). Such alarm indicator **34** can facilitate feedback and/or machine learning at alarm management engine **12**.

(50) In a general sense, alert management engine **12** of system **10** configured to perform various steps. First, alert management engine **12** receives status data **27** corresponding to a user, where status data **27** modifies at least one user fatigue model attribute **26**. Second, alert management engine **12** updates corresponding user fatigue model **24** based on status data **27**. Third, alert management engine **12** determines an alarm fatigue level based on updated user fatigue model **24**.

Fourth, alert management engine **12** generates alarm condition **44** targeting the user based on the calculated alarm fatigue level (among other parameters). Finally, adapter **46** modifies alarms generated at medical device **16** according to alarm condition **44** and distributes modified alarm **78** to the user.

(51) In various embodiments, system **10** may facilitate aggregation and/or management of alarms in a consumable fashion. In some embodiments, adapter **46** associated with medical device **16** may comprise an alarm monitor, an alarm filter, an alarm modifier, a memory element for storing data, and a processor that executes instructions associated with the data. The processor and the memory element cooperate such that adapter **46** is configured for receiving alarm condition **44** from alarm management engine **12**, receiving an alarm from medical device **16**, modifying the alarm according to alarm condition **44** and generating alarm indicator **34** based on the modification. In some embodiments, alarm indicator **34** comprises a feedback to alarm management engine **12**, and alarm condition **44** is further based on the feedback. Alarm condition **44** may be configured to increase a likelihood of the user responding to the modified alarm.

(52) In some embodiments, alarm condition **44** is based on an alarm fatigue level of a user of medical device **16**, the alarm fatigue level being based at least on medical device model **28** and patient condition **32**. The alarm may be based on a physiological measurement of a patient by medical device **16**. In some embodiments, modifying the alarm can include deleting the alarm based on alarm condition **44**. In some other embodiments, modifying the alarm can include changing a format of the alarm. In yet other embodiments, modifying the alarm can include changing a distribution mode of the alarm.

(53) Turning to the infrastructure of system **10**, alarm management engine **12** can be embodied as computer executable instructions stored on one or more non-transitory computer-readable media (e.g. hard drives, optical storage media, flash drives, ROM, RAM, etc.) that, when executed by one or more processors, cause the processors to execute the functions and processes described herein. In some embodiments, alarm management engine **12** may execute in a distributed manner, portions of which are integrated into different adapters **46**. For example, a portion of alarm management engine **12** that can calculate alarm condition **44** for medical device A may be integrated into adapter **46** corresponding to medical device A; another portion of alarm management engine **12** that can calculate alarm condition **44** for medical device B may be integrated into adapter **46** corresponding to medical device B; and so on. In other embodiments, alarm management engine **12** may execute in a centralized manner, calculating alarm condition **44** for a plurality of medical devices **16** and communicating alarm condition **44** over network **11** to relevant medical devices **16**.

(54) In some embodiments, alarm management engine **12** can be integrated into a single computing device or distributed among a plurality of computing devices (either locally or remotely located from one another) communicatively coupled via data exchange interfaces (e.g., short-range, long-range, wireless, wired, near-field communication, Bluetooth, Ethernet, Wi-Fi, USB, etc.), and/or connected via local or long-range networks (e.g. Internet, cellular, local-area networks, wide-area networks, intranets, etc.). In some embodiments, alarm management engine **12** can be embodied as one or more dedicated hardware computing devices specifically programmed (e.g. via firmware) to execute the functions and processes described herein.

(55) In some embodiments, alarm management engine **12** can be incorporated into existing alarm management systems via installation of appropriate hardware and/or software updates associated with the functions described herein. For example, one suitable alarm management system is NantHealth's Magellan™ or cOS™ system, including functionalities as described herein.

(56) Turning to FIG. 2, FIG. 2 is a simplified block diagram illustrating network **11** according to an example embodiment of system **10**. Alarm management engine **12** may execute in a server **50** connected over a local area network **52** with medical device **16** and user device **14**. Server **50** may connect over Internet **54** to a clinical operating system (cOS) **56** that may communicate with user fatigue model database **18**, medical device model database **20** and patient conditions database **22**.

Note that LAN 52 and Internet 54 may be connected over a router (not shown) situated at an edge of LAN 52. In some embodiments, LAN 52 may comprise a portion of an enterprise network.

(57) In various embodiments, cOS 56 integrates clinical, financial, operational and environmental data into a single platform. In various embodiments, cOS 56 comprises a cloud-based platform for patient records, medical devices, imaging systems, financial systems, costing systems, evidence-based clinical pathways, and personalized genomic and proteomic data. cOS 56 combines and organizes pertinent medical information for easy access and utilization at the point of care. In various embodiments, alarm management engine 12 may connect to cOS 56 over Internet 54, and access user fatigue model 24, patient condition 32, medical device model 28, etc. through cOS 56. cOS 56 can facilitate privacy of health records, appropriate authentication, and other security measures consistent with various health care privacy related laws and regulations. For example, alarm management engine 12 may establish a secure tunnel with cOS 56 to access user fatigue model database 18, medical device model database 20 and patient conditions database 22.

(58) In some embodiments, server 50 may comprise a physical computing device, such as a desktop server. In other embodiments, server 50 may execute in a rack server, for example, in a computing station located remotely from medical device 16 and user device 14, but nevertheless in communication with medical device 16 and user device 15 over LAN 52. LAN 52 may include switches and other network elements that can facilitate wired or wireless communication between medical device 16, user device 14 and alarm management engine 12.

(59) Turning to FIG. 3, FIG. 3 is a simplified block diagram illustrating network 11 according to another example embodiment of system 10. Alarm management engine 12 may execute in a server 50 connected over a first local area network 52 (LAN A) with medical device 16 and user device 14. Server 50 may connect over a second local area network 52 (LAN B) to user fatigue model database 18, medical device model database 20 and patient conditions database 22. In some embodiments, LAN A and LAN B may comprise the same LAN; in other embodiments, LAN A and LAN B may comprise separate portions of an enterprise network; in yet another embodiment, LAN A and LAN B may comprise separate enterprise networks linked together by appropriate secure network elements, such as switches and routers; in yet other embodiments, LAN A and LAN B may comprise virtual local area networks (VLANS) of an enterprise network.

(60) Note that various implementations of LAN A and LAN B may be encompassed by the broad scope of the embodiments. In one example embodiment, LAN A and LAN B may be connected over a wireless network. In another embodiment, LAN A and LAN B may be connected over the Internet (or other WAN), but through secure tunnels and/or other network security measures tailored to transparently and seamlessly cause LAN B to appear to belong to the same network as LAN A. For example, user fatigue model database 18, medical device model database 20 and patient conditions database 22 may be stored and/or implemented in LAN B, comprising a data center network geographically or physically separate from LAN B, comprising a hospital network, but nevertheless connected to LAN A through appropriate communication channels.

(61) Turning to FIG. 4, FIG. 4 is a simplified block diagram illustrating network 11 according to yet another example embodiment of system 10. Alarm management engine 12 may communicate with medical device 16 and user device 14 across Internet 54 through cOS 56. For example, cOS 56 may execute in a cloud network and receive status data 27, alarm indicator 34, etc. from medical device 16 and user device 14, communicate them with alarm management engine 12, and facilitate access by alarm management engine 12 of user fatigue model database 18, medical device model database 20 and patient conditions database 22 stored/implemented in the cloud. In some embodiments, alarm management engine 12 may execute in a server communicatively coupled to cOS 56; in other embodiments, alarm management engine 12 may execute in a same server as cOS 56. In some embodiments, user fatigue model database 18, medical device model database 20, and patient conditions database 22 may be implemented in a separate storage area network (SAN) that is communicatively coupled to cOS 56 and/or alarm management engine 12.

(62) Turning to FIG. 5, FIG. 5 is a simplified block diagram illustrating another example embodiment of system **10**. In some embodiments, alarm condition **44** may be provided based on subscriptions to an alarm service. A subscriptions database **58** may be communicatively coupled to alarm management engine **12**. Subscriptions database **58** may include a medical device group-subscription level table **60** and subscription levels **62**. A plurality of medical device **16** may be grouped into separate medical device groups **64**. Alarm management engine **12** may access medical device group-subscription level table **60** in subscriptions level database **58** and generate alarm condition **44** based on subscription levels **62** (in addition to other parameters as already discussed herein).

(63) For example, medical devices C and D may be grouped into medical device group A; medical devices E and F may be grouped into medical device group B. Medical device groups **64** may be classified (e.g., grouped) according to any suitable grouping attribute. For example, the grouping attribute may comprise location of medical device **16**: medical device group A may correspond to medical devices located in an ICU of a hospital; medical device group B may correspond to medical devices located in an outpatient clinic of the hospital. In another example, the grouping attribute may comprise device types: medical device group A may correspond to heart monitors; medical device group B may correspond to blood pressure monitors. In yet another example, the grouping attribute may comprise departments: medical device group A may comprise medical devices belonging to the oncology department; medical device group B may comprise medical devices belonging to the emergency room department. Virtually any suitable grouping attribute may be included within the broad scope of the embodiments.

(64) Each medical device group **64** may subscribe to different subscription levels **62**. Each subscription level **62** may specify certain types of alarm condition **44**; certain types of analyses used to generate alarm condition **44**; certain attributes (e.g., parameters, variables, etc.) to be taken into consideration to compute alarm condition **44**; etc. Virtually any suitable differentiator of alarm condition **44** may be included in subscription level **62** within the broad scope of the embodiments.

(65) During operation, assume, merely for example purposes and not as a limitation, that alarm management engine **12** generates alarm condition O for identical medical devices C and E based on alarm fatigue levels, medical device model **28**, patient condition **32**, etc., as described herein. Assume that alarm condition **0** indicates that alarms are to be generated for users P and Q only when a patient's blood pressure crosses 130/90 mm Hg.

(66) Alarm management engine **12** may access subscriptions database **58** to determine subscription level **62** corresponding to medical devices C and E. For example, alarm management engine **12** may determine (e.g., from preconfigured information, polling medical devices C and E, or other appropriate methods) that medical device C belongs to medical device group A and medical device E belongs to medical device group B. From medical device group-subscription level table **60**, alarm management engine **12** may determine that medical device group A subscribes to subscription level G, which indicates that alarm signals be communicated in a vibratory mode to specific user devices such as watches (according to subscription level **62**) and medical level group B subscribes to subscription level H, which indicates that alarm signals be communicated audibly and visually on the medical device itself.

(67) Alarm management engine **12** may modify alarm condition O to alarm condition **1** for medical device C based on information from subscriptions database **58**; and modify alarm condition O to alarm condition **2** for medical device E. Thus, alarm condition **1** may indicate that alarms are to be generated for users P and Q only when the patient's blood pressure crosses 130/90 mm Hg and the users are to be notified via vibratory mode to specific user devices; alarm condition **2** may indicate that alarms are to be generated for users P and Q only when the patient's blood pressure crosses 130/90 mm Hg and the alarms are to be audibly and visually activated at medical device E.

(68) Turning to FIG. 6, FIG. 6 is a simplified block diagram illustrating example details according to an example embodiment of system **10**. Adapter **46** may be communicatively coupled to medical

device **16** and alarm management engine **12**. In some embodiments, adapter **46** may be remote from medical device **16** and communicate wirelessly with medical device **16**. In other embodiments, adapter **46** may be detachably attached (e.g., through universal serial bus (USB) or other such electrical connectors) to medical device **16**. In yet other embodiments, adapter **46** may be permanently attached to medical device **16**. In yet other embodiments, adapter **46** may comprise a separate physical appliance (e.g., component) embedded inside a chassis of medical device **16** and integrated therein. In yet other embodiments, adapter **46** may comprise an inseparable component integrated into a hardware of medical device **16**; for example medical device **16** could integrate a Digi Connect ME® device that has been configured or programmed according to the disclosed techniques.

(69) In yet other embodiments, adapter **46** may comprise a software application executing inside medical device **16**. In yet other embodiments, adapter **46** may comprise a software application executing in a computing device separate from medical device **16** and communicatively coupled thereto (e.g., connected via Ethernet links, connected through a controller, etc.). For example, adapter **46** may comprise a software application (e.g., software container, object, module, etc.) executing alongside (e.g., with, concurrently, etc.) alarm management engine **12** in the same server.

(70) Adapter **46** may include a processor **66**, a memory element **68**, an alarm monitor **70**, and an alarm filter **72**, an alarm modifier **74**. Adapter **46** may receive alarm condition **44** from alarm management engine **12**. Alarm condition **44** may be based on an alarm fatigue level of a user of medical device **16**, the alarm fatigue level being based on (among other parameters), user fatigue model **24**, medical device model **28** and patient condition **32**.

(71) Adapter **46** may also receive one or more alarms **76** generated by medical device **16**. In some embodiments, alarm **76** can comprise electrical signals; in other embodiments, alarm **76** can comprise visual, auditory, vibratory and other kinds of signals; in yet other embodiments, alarm **76** can comprise a signal indicative of an alarm; in yet other embodiments, alarm **76** comprises signals instructing an alarm to be generated. Alarm monitor **70** at adapter **46** is configured to receive and identify alarms **76**. For example, alarm **76** comprises an electrical signal and alarm monitor **70** may comprise an electrical component configured to change its property upon receipt of the electrical signal. In another example, alarm **76** comprises an audible signal, and alarm monitor **70** may comprise a microphone sufficiently sensitive to pick up the audible signal. In yet another example, alarm signal comprises an optical signal, and alarm monitor **70** may comprise a light sensitive component that may activate upon receipt of the optical signal. Virtually any suitable type of alarm monitor **70** that can interface with alarms **76** may be used within the broad scope of the embodiments.

(72) In various embodiments, adapter **46** modifies alarms **76** to modified alarm **78** according to alarm condition **44**. Note that alarm condition **44** may be configured to increase a likelihood of the user (of medical device **16**) responding to modified alarm **78** (e.g., relative to a likelihood of the user responding to alarms **76**). Modification may comprise filtering alarms **76** to delete some of alarms **76** based on alarm condition **44** (e.g., alarm condition **44** may specific alarms once every 10 minutes so that any alarms generated in the intervening 9 minutes are discarded); modification may comprise altering a format of alarms **76** (e.g., changing a vibratory format to a visually blinking format; changing a text message alarm to a beeping sound alarm; changing alarm **76** comprising an alarm instruction to a vibration on the user's wearable device; etc.). In some embodiments, modification may comprise changing a distribution mode of alarms **76**. For example, alarms **76** may comprise a blinking light on medical device **16**; modified alarm **78** may comprise a text message on cell phones of a plurality of registered users.

(73) In some embodiments, modification of alarms **76** may be accomplished by alarm filter **72** and alarm modifier **74**. Alarm filter **72** is configured to filter alarms **76** according to alarm condition **44**. For example, medical device **16** may generate alarms **76** indiscriminately whenever certain manufacturer preconfigured alert conditions are met. On the other hand, alarm condition **44** may be

tailored to specific users, environments, etc. taking into consideration alarm fatigue levels, and other factors as described herein. Accordingly, alarm filter **72** may reduce the number of alarms; change the type of alarms; or otherwise generate instructions to change alarms **76** in a suitable manner according to alarm condition **44**. In some embodiments, alarm filter **72** may take as inputs signals corresponding to alarms **76** from alarm monitor **70** and alarm condition **44** from alarm management engine **12** and produce as output instructions to generate modified alarm **78**.

(74) Alarm modifier **74** may generate modified alarm **78** according to the instructions from alarm filter **72**. In some embodiments, each modified alarm **78** may also cause generation of alarm indicator **34**, which may be sent back to alarm management engine **12** (e.g., as feedback). In some embodiments, alarm indicator **34** may be used to compute alarm condition **44**.

(75) Turning to FIG. 7, FIG. 7 is a simplified flow diagram illustrating example operations **80** that may be associated with adapter **46** according to an embodiment of system **10**. At **82**, adapter **46** receives alarm condition **44** from alarm management engine **12**. In some embodiments, alarm condition **44** is based on an alarm fatigue level of a user of medical device **16**, the alarm fatigue level based on at least user fatigue model **24**, medical device model **28**, and patient condition **32**. At **84**, adapter **46** receives alarm **76** from medical device **16**. At **86**, adapter **46** modifies alarm **76** to generate modified alarm **78**. In various embodiments, alarm condition **44** is configured to increase a likelihood of the user responding to modified alarm **78**. At **88**, adapter **88** generates alarm indicator **34** based on the modification.

(76) Turning to FIG. 8, FIG. 8 is a simplified block diagram illustrating example details of a system **100** for alarm fatigue management. System **100** includes two components: a database **102**, which can be stored either locally or remotely, and an alarm management engine **116**. Database **102** can include N user fatigue models **104**, **106**, **108**, where N is an integer value corresponding to the number of users within system **100**. Each user fatigue model can include one or more user fatigue model attributes. User fatigue model **104** holds, for example, M user fatigue model attributes **110**, **112**, and **114**, where M is an integer value corresponding to the number of attributes held in the user fatigue model.

(77) Alarm management engine **116** can receive alarm indications via input **118**, and it can additionally communicate with J number of user devices where J is an integer value corresponding to the number of user devices used with system **100**.

(78) Alarm management engine **116** can be embodied as computer executable instructions stored on one or more computer-readable media (e.g. hard drives, optical storage media, flash drives, ROM, RAM, etc.) that, when executed by one or more processors, cause the processors to execute the functions and processes of the inventive subject matter. In some embodiments, alarm management engine **116** can be integrated into a single computing device or distributed among a plurality of computing devices (either locally or remotely located from one another) communicatively coupled via data exchange interfaces (e.g., short-range, long-range, wireless, wired, near-field communication, Bluetooth, Ethernet, Wi-Fi, USB, etc.), connected via local or long-range networks (e.g. Internet, cellular, local-area networks, wide-area networks, intranets, etc.). In some embodiments, alarm management engine **116** can be embodied as one or more dedicated hardware computing devices specifically programmed (e.g. via firmware) to execute the functions and processes of the inventive subject matter.

(79) In some embodiments, alarm management engine **116** can be incorporated into existing alarm management systems via the installation of required hardware and/or software updates associated with the functions of the inventive subject matter. For example, one suitable alarm management system is NantHealth's Magellan<sup>TM</sup> of cOS<sup>TM</sup> system.

(80) User fatigue model attributes can be considered to be factors or characteristics that can affect a user's fatigue level. User fatigue model attributes can include many different types of information, including information conveyed via the status data and information that is set based on details about a particular user. These attributes can be changed and/or updated periodically or

continuously, depending on the sampling frequency in an embodiment. Additionally, the attributes can contain historical data rather than repeatedly updating a single piece of information.

(81) Some examples of user fatigue model attributes include personality factors and physiological factors. Personality factors are factors that might indicate a person is more or less prone to alarm fatigue (e.g., irritability, psychological disorders, and personality type). Physiological factors can include factors that pertain to a certain user's physical abilities (e.g., deafness, blindness, color blindness, sensitivity to particular light and/or sound patterns, and varying degrees of disabilities related to these conditions and others).

(82) In some embodiments, system **100** comprising database **102** and alert management engine **116** can modify operating parameters of medical devices to help reduce alarm fatigue at the sensor level. In this aspect, each user fatigue model **104-108** is associated with a corresponding user, and alert management engine **116** is configured to perform the following steps. It receives user status data that includes at least one status attribute. It receives patient status data from one or more medical devices where the medical devices have at least one an alarm condition parameter. It updates the first user's user fatigue model based on the first user's status data. It determines an alarm fatigue level based on the updated first user fatigue model. It modifies the at least one operational parameter of the at least one medical device based on at least the alarm fatigue level of the user.

(83) Turning to FIG. **9**, FIG. **9** is a simplified flow diagram illustrating example operations **200** that may be associated with alarm management engine **116** according to some embodiments of system **100**. In step **202**, alarm management engine **116** receives status data. Status data collected in step **202** can take many different forms, including physiological data, user shift data, and environmental alarm data. In step **204**, a user fatigue model is updated based on the status data received in step **202**. Multiple user fatigue models can be updated simultaneously or in sequence as the data arrives—the system is limited in this respect by the number of users accounted for in the system.

Additionally, incoming status data can either overwrite previously stored status data within a user fatigue model or it can be stored alongside all previously stored status data to create a historical set of status data. In step **206**, an alarm fatigue level is determined based on the previously updated user fatigue model of step **204**. In step **208**, system **100** receives an indication of a medical alert requiring a response from a user. The system can receive inputs from medical sensing equipment that collect physiological information from a patient. When, for example, a medical sensor indicates a heart rate has flat lined, an alert condition is met. In step **210**, system **100** generates an alarm directed to a user taking into account that user's alarm fatigue level. In step **212**, system **100** distributes the generated alarm based on the alarm fatigue level of step **206** and the indication of an alert of step **208**.

(84) Turning to FIG. **10**, FIG. **10** is a simplified diagram illustrating example details and operations that may be associated with embodiments of system **100**. An alarm management engine **300** performs step **306** by determining alarm fatigue based on status data collected from one or more medical devices **302** and performs step **308** by adjusting alarm conditions of one or more medical devices **302** based on the one or more user fatigue levels. In some embodiments, instead of simply altering alarms based on user fatigue levels, alarm management engine **300** is configured to alter alarm conditions of one or more medical devices **302**. An advantage of altering alarm conditions of a medical device is that each individual device **310, 312, 314, 316** can be modified to indicate an alarm based on one or more nonstandard operational parameters. For example, if a particular medical device is prone to issuing excessive alarms because a threshold for meeting an alarm condition is too low (e.g., when a heart rate drops below 50 BPM, an alarm condition is met), the threshold comprising the alarm condition can be altered (e.g., such that the alarm condition is met only when the heart rate drops below 30 BPM), thus reducing a number of alarms generated by that particular medical device.

(85) In some embodiments, the alarm condition can comprise a threshold value for the patient

status data, for example, above which the medical device generates an alarm. For example, if the medical device is a heart rate monitor and it is originally configured to generate an alarm if a patient's heart rate drops below 50 BPM, then system **100** can modify the medical device to generate the alarm only when that patient's heart rate drops below 30 BPM.

(86) In some embodiments, the alarm condition can comprise a maximum frequency at which the medical device can generate an alarm (or indication of an alert). For medical devices that detect physiological information that changes very slowly, it can be advantageous to permit the medical device to only generate an alarm every so often. For example, if a patient's body temperature drops below a threshold where the medical device will generate an alarm, the slow changing nature of body temperature may result in fluctuation about that threshold level. These fluctuations can in turn cause the medical device to generate many alarms very quickly as the patient's body temperature changes right around the threshold temperature. System **100** can remedy this by altering the alarm condition of the medical device such that it can generate an alarm only once every 10 minutes, which would allow the patient's body temperature to change without generating unnecessary alarms.

(87) In some embodiments, the alarm condition can comprise a sampling frequency of a medical device. Thus, changing the alarm condition can change how frequently the medical device gathers physiological information from a patient. This can be advantageous in that it can reduce the number of alarms the medical device generates.

(88) In some embodiments, the alarm condition can comprise measurement sensitivity, which can determine a minimum amount of change a physiological measurement undergoes before the medical device will detect that change. This can also help to reduce unnecessary alarm generation. In an example, if a medical device is set to detect body temperature changes only when the temperature change is 0.2 degrees or greater, fewer alarms would be generated based on small fluctuations around the threshold value.

(89) The same concepts can be extended to any type of medical device. Medical devices can include devices that directly measure physiological information, or machines that conglomerate that type of information (e.g., Phillips IntelliVue Patient Monitor and other comparable systems known in the art). Physiological quantities that can be measured by medical devices include: monitoring for cardiac problems such as arrhythmias (e.g., atrial, fibrillation, ventricular fibrillation, ventricular tachycardia, and premature ventricular contractions) and asystole (i.e., lack of measurable electrical cardiac activity); monitoring for pulse rate problems such as tachycardia and bradycardia; monitoring for respiration problems (for example, via oximetry) such as apnea, tachypnea, and desaturation; and monitoring for blood pressure problems such as hypertension and hypotension. Alarm conditions can correspond to a particular ECG waveform, or to an elevated blood toxin level in some embodiments. In some embodiments, alarms can include the physiological measurements thereby conveying information to the recipient while simultaneously alerting the recipient that a response is required.

(90) Several medical devices may operate concurrently in a Medical Intensive Care Unit (MICU) and Coronary Care Unit (CCU). For example, a ventilator and Swan-Ganz catheter (a flexible tube inserted through the right side of the heart that monitors pressures and outflow from the heart) can both generate alarms when the devices measure status data that meets respective internal alarm conditions. Any medical device connected to alarm management engine **300** can be modified suitably to change the alarm condition within the medical device itself. The result is a reduced number of alarms in the course of normal operation of system **100**, thereby decreasing overall alarm fatigue of one or more system users.

(91) Alternatively, alarms can be distributed over a centralized system, for example, via alarm devices that are fixed in place throughout a hospital. The alarm devices can be capable of projecting lights of different colors and intensities, as well as sounds of different frequencies and intensities. One possible alarm device comprises a monitor such as a television with accompanying

speakers where a distributed alarm can be presented as text, light, and/or sound. In some embodiments, system **100** can operate using an existing alarm system. System **100** can use the alarm distribution capabilities of the existing system to distribute its alarms. Regardless of the distribution method, in one way or another distribution of alarms will likely affect users beyond merely the target user. However, since the system is continually (or continuously) collecting status data from a plurality of users, these effects are taken into account in terms of each user's fatigue level.

(92) Some embodiments of alarm management system **100** can operate in real-time, monitoring relevant user activities as they occur and using the monitoring information to adjust its output. To do so, system **100** employs a feedback loop. Status data updates a profile, which is used to determine alarm fatigue level to generate an output, which in turn influences the alarm fatigue level. For example, as a user experiences more and more alarms throughout a shift, that user's alarm fatigue rises. System **100**, sensing and quantifying a rise in alarm fatigue level, modulates its alarm outputs to that user to mitigate the effects of that user's alarm fatigue.

(93) Thus, a closed-loop control system (e.g., a proportional-integral-derivative control system) can be implemented to manage changes in alarm outputs. Closed-loop control systems can be under-damped, over-damped, or critically damped. System **100** can be configured to account for any hysteresis of its feedback loop, where hysteresis indicates a dependence of the output of system **100** on its current input and its history of past inputs. In other words, system **100** may compensate for possible staleness of data when contemplating how quickly that data can be used to influence output. To this end, system **100** can maintain frames of reference in its database for alarm fatigue levels that can span hours, days, weeks, months, and/or years.

(94) One should also appreciate that system **100** is also able to monitor trends in available data in order to trigger shifts in how alarms are to be presented. The trends could be with respect to alarm data, user behavior data, or other types of data. System **100** can, in some embodiments, represent the trends as derivatives or rates of change of one value versus a second value;  $dx/dy$  for example. It should be appreciated that the derivatives or rates can be empirically measured across appropriate corresponding units of measure. For example, the rate of change could be missed alarms per day where number of alarms missed could be measured as a difference from one day to the next. More interesting examples might include a derivative associated with other values besides time. As an example, consider a scenario where the severity of an alarm ( $S$ ) might correlated with number of missed alarms ( $M$ ) for a specific individual. In such a scenario,  $dS/dM$  could be indicative of how well a user deals with alarm severity and that the user's a fatigue model should include triggers for such measures.

(95) Beyond merely measuring trends as first order derivatives, it should be appreciated the trends could also include higher order derivatives of the available, measurable metrics. As system **100** collects data over time, system **100** can monitor the higher order derivatives to determine if one or more users are becoming fatigue at a faster rate for example. Returning to the example of missed alarms per unit time ( $dM/dt$ ), system **100** can be configured to accept a constant rate at which the alarms are missed, at least to with a threshold. Note that when  $dM/dt$  is constant, its higher derivatives are zero;  $d^2M/dt^2=0$ . However, should the higher order derivatives shift to unacceptable non-zero values (e.g., much greater than zero), then the system can begin altering its alarm strategy to address the possible indication that the user has become fatigued. The reader is reminded that the technique can also be leveraged for higher order rates of changes that do not involve time;  $d^2x/dy^2$ ,  $d^3x/dy^3$ ,  $d^4x/dy^4$ , etc.

(96) In some embodiments, system **100** can generate recommendations as to what actions a user can/should take to reduce alarm fatigue levels (e.g., take a day off or generate a new work schedule). However, there is a difference between what system **100** would recommend versus what might be practical. For example, if a floor is short staffed, system **100** may be configured to consider a certain alarm fatigue level that is different from the alarm fatigue level when the floor is

not short-staffed. In some embodiments, system **100** can recommend adjustments to shift schedules. In other embodiments, system **100** can be configured to override its normal functions in the event of an emergency (e.g., an all-hands-on-deck emergency where maximum alert intensity is needed to get the attention of all relevant personnel).

(97) To the extent that it can influence an ecosystem, system **100** may likely have to compensate for a certain level of fatigue and/or risk. In other words, external factors influencing alarm fatigue levels may be present that are unaccounted in system **100**. While system **100** can monitor its users to minimize those impacts, risk can likely never be reduced to zero and so system **100** can take that risk into account when determining alarm conditions.

(98) In some embodiments, alarm management engine **300** additionally: (1) determines an alarm fatigue level delta corresponding to a difference between at least two consecutive alarm fatigue levels determined during a user's shift; (2) analyzes the status data associated with each user fatigue model update used for each of the at least two consecutive alarm fatigue levels determined during the at least one historical user's shift to determine at least one principal contributor to the at least one alarm fatigue level delta (e.g., via, for example, principal component analysis or eigenvalue decomposition it determines the causes of alarm fatigue by isolating the largest contributing factors); (3) receives an indication of a medical alert requiring a response from the first user; and (4) generates an alarm targeting the first user based on the alarm fatigue level, the at least one principal contributor, and the indication of an alert (i.e., to try to prevent alarm fatigue before it happens by addressing a particular person's own largest causes).

(99) System **100** can exist within various operating system environments, such as cOS™. The cOS™ clinical operating system is designed to bring together clinical, financial, operational and environmental data to identify and solve complex healthcare problems such as quality, cost, and outcomes at a health system level, hospital level, service line level, physician level and the patient level. The cOS architecture is designed using principles of big data, enterprise data warehouse, and big memory that allow for proactive decision support as opposed to retrospective business intelligence. Proactive decision support allows health system executives to predict and solve problems before they occur (e.g., solve alarm fatigue problems proactively rather than retroactively). The cOS™ platform allows innovation to thrive because it is a completely open system with documented web services that can be leveraged by the health system as well as third party software development companies to build new and innovative applications.

(100) Within the cOS™ system, system **100** can be implemented and can execute at various levels. For example, at a hospital level, system **100** can manage alarm fatigue levels of hospital personnel and staff. At a physician level, system **100** can be directed towards medical doctors currently working a shift. At a patient level, system **100** can be implemented to manage alarm fatigue levels of individuals working with a particular patient.

(101) Some embodiments of system **100** are capable of reporting performance statistics (e.g., for JACHO) to verify that alarm fatigue levels have been reduced and that environments where system **100** has been deployed are safer than national average (e.g., system **100** has resulted in reduced overall alarm fatigue levels compared to the national average resulting in better patient care).

(102) In various embodiments, system **100** can be applied to any type of scenario that requires attention of a user. One such application can include notifications within a computer operating system. The system could monitor user responses and determine when an alarm should be altered to better capture the attention of a user experiencing alarm fatigue. For example, scheduling software can provide notifications to remind a user of important meetings, but each of those notifications is identical and when a user has a very busy schedule that user may become desensitized to the alerts. The alarm management system can be advantageously applied to such a system to help the user from missing important meetings or deadlines.

(103) In addition to offices and hospitals, system **100** can also be implemented in fire stations, police stations, and/or military bases, industrial settings, gaming environments, or in other

industries where alarms and alerts are used heavily. For example, in a fire station, fire fighters may be subjected to many alarms per day or per shift, which could lead to alarm fatigue if nothing is done to prevent the condition from developing. The same holds true for police stations. A consequence of alarm fatigue in these situations may be delayed response times to an emergency such as a fire or a burglary. On military bases, soldiers are exposed to many different alarms that provide different types of information. Developing alarm fatigue in this setting can lead to missed alarms, and in a scenario where, for example, an alarm is sounded in a war-zone, missing that alarm can mean the difference between life and death. In industrial settings, equipment and other alarms blaring over the course of a day can result in decreased concentration of workers on their respective tasks at hand. In the operation of heavy machinery, performing tasks requiring high levels of precision, and/or handling of dangerous materials, a lapse in concentration can lead to an accident causing injuries, deaths, damage to equipment, and loss of progress. In gaming settings, an ability to perform at a high level and an immersion in the game can both be critical to a player's enjoyment of the game, which can be disrupted by an excess of warnings and/or alarms.

(104) It should be apparent to those skilled in the art that many more modifications besides those already described are possible without departing from the inventive concepts disclosed herein. Moreover, in interpreting both the specification and the claims, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refers to at least one of something selected from the group consisting of A, B, C . . . and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc.

(105) The foregoing discussion provides many example embodiments of systems and methods for alarm fatigue management. Although each embodiment represents a single combination of various elements, all possible combinations of the disclosed elements are intended to be included in the broad scope of the disclosure. Thus if one embodiment comprises elements A, B, and C, and a second embodiment comprises elements Band D, then the scope of the disclosure is considered to include other remaining combinations of A, B, C, or D, even if not explicitly disclosed.

(106) As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

(107) Unless the context dictates the contrary, all ranges set forth herein should be interpreted as being inclusive of their endpoints and open-ended ranges should be interpreted to include only commercially practical values. Similarly, all lists of values should be considered as inclusive of intermediate values unless the context indicates the contrary. Note that any recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein.

(108) Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member can be referred to and claimed individually or in any combination with other members of the group or other elements found herein. One or more members of a group can be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

(109) Note that in this Specification, references to various features (e.g., elements, structures, modules, components, steps, operations, characteristics, etc.) included in “one embodiment”,

“example embodiment”, “an embodiment”, “another embodiment”, “some embodiments”, “various embodiments”, “other embodiments”, “alternative embodiment”, and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments. The use of any and all examples, or exemplary language (e.g. “such as”) provided with respect to certain embodiments herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the embodiments otherwise claimed. No language in the specification should be construed as indicating any non-claimed essential.

(110) In example implementations, at least some portions of the activities outlined herein may be implemented in software in, for example, alarm management engine **12**. In some embodiments, one or more of these features may be implemented in hardware, provided external to these elements, or consolidated in any appropriate manner to achieve the intended functionality. The various network elements may include software (or reciprocating software) that can coordinate in order to achieve the operations as outlined herein. In still other embodiments, these elements may include any suitable algorithms, hardware, software, components, modules, interfaces, or objects that facilitate the operations thereof.

(111) Furthermore, alarm management engine **12** and various other components described and shown herein (and/or its associated structures) may also include suitable interfaces for receiving, transmitting, and/or otherwise communicating data or information in a network environment. Additionally, some of the processors and memory elements associated with the various nodes may be removed, or otherwise consolidated such that a single processor and a single memory element are responsible for certain activities. In a general sense, the arrangements depicted in the FIGURES may be more logical in their representations, whereas a physical architecture may include various permutations, combinations, and/or hybrids of these elements. It is imperative to note that countless possible design configurations can be used to achieve the operational objectives outlined here. Accordingly, the associated infrastructure has a myriad of substitute arrangements, design choices, device possibilities, hardware configurations, software implementations, equipment options, etc. Moreover, all methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context.

(112) As used herein, and unless the context dictates otherwise, the term “coupled to” is intended to include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements). Therefore, the terms “coupled to” and “coupled with” are used synonymously.

(113) In some of example embodiments, one or more memory elements (e.g., memory element **38**, **68**) can store data used for the operations described herein. This includes the memory element being able to store instructions (e.g., software, logic, code, etc.) in non-transitory media such that the instructions are executed to carry out the activities described in this Specification. These devices may further keep information in any suitable type of non-transitory storage medium (e.g., random access memory (RAM), read only memory (ROM), field programmable gate array (FPGA), erasable programmable read only memory (EPROM), EEPROM, etc., software, hardware, or in any other suitable component, device, element, or object where appropriate and based on particular needs.

(114) A processor can execute any type of instructions associated with the data to achieve the operations detailed herein in this Specification. In one example, processors (e.g., processor **36**, **66**) could transform an element or an article (e.g., data) from one state or thing to another state or thing. In another example, the activities outlined herein may be implemented with fixed logic or programmable logic (e.g., software/computer instructions executed by a processor) and the elements identified herein could be some type of a programmable processor, programmable digital logic (e.g., a field programmable gate array (FPGA), an erasable programmable read only memory (EPROM), an electrically erasable programmable read only memory (EEPROM)), an ASIC that

includes digital logic, software, code, electronic instructions, flash memory, optical disks, CD-ROMs, DVD ROMs, magnetic or optical cards, other types of machine-readable mediums suitable for storing electronic instructions, or any suitable combination thereof.

(115) The information being tracked, sent, received, or stored in system **10** could be provided in any database, register, table, cache, queue, control list, or storage structure, based on particular needs and implementations, all of which could be referenced in any suitable timeframe. Any of the memory items discussed herein should be construed as being encompassed within the broad term ‘memory element.’ Similarly, any of the potential processing elements, modules, and machines described in this Specification should be construed as being encompassed within the broad term ‘processor.’

(116) It is also important to note that the operations and steps described with reference to the preceding FIGURES illustrate only some of the possible scenarios that may be executed by, or within, the system. Some of these operations may be deleted or removed where appropriate, or these steps may be modified or changed considerably without departing from the scope of the discussed concepts. In addition, the timing of these operations may be altered considerably and still achieve the results taught in this disclosure. The preceding operational flows have been offered for purposes of example and discussion. Substantial flexibility is provided by the system in that any suitable arrangements, chronologies, configurations, and timing mechanisms may be provided without departing from the teachings of the discussed concepts.

(117) Note also that the disclosed subject matter herein enables construction or configuration of a medical device to operate on digital data (e.g., raw sensor data, alarm condition, etc.), beyond the capabilities of a human or un-configured (e.g., off-the-shelf) medical device. Although the digital data represents alarm conditions, it should be appreciated that the digital data is a representation of one or more digital models of alarm conditions and not the actual alarms themselves, which comprise activities or operations performed by the medical device and/or adapters. By instantiation of such digital models in the memory of the medical device (and/or adapter), the medical device (and/or adapter) is able to manage the digital models in a manner that could provide utility to an individual (e.g., a user of the system) that the individual would lack without such a tool.

(118) It should also be noted that any language directed to a computer should be read to include any suitable combination of computing devices, including servers, interfaces, systems, databases, agents, peers, engines, controllers, or other types of computing devices operating individually or collectively. One should appreciate the computing devices comprise a processor configured to execute software instructions stored on a tangible, non-transitory computer readable storage medium (e.g., hard drive, solid state drive, random access memory (RAM), flash memory, read-only memory (ROM), etc.). The software instructions can configure a suitable computing device to provide the roles, responsibilities, or other functionality as discussed herein with respect to the disclosed apparatus. In some embodiments, the various servers, systems, databases, or interfaces exchange data using standardized protocols or algorithms, possibly based on hyper-text transfer protocol (HTTP), hyper-text transfer protocol secure (HTTPS), Advanced Encryption Standard (AES), public-private key exchanges, web service application programming interfaces (APIs), known financial transaction protocols, or other electronic information exchanging methods. Data exchanges preferably are conducted over a packet-switched network, the Internet, local area network (LAN), wide area network (WAN), virtual private network (VPN), or other type of packet switched network.

(119) As used in the description herein and throughout the claims that follow, when a system, engine, server, device, module, or other computing element is described as configured to perform or execute functions on data in a memory, the meaning of “configured to” or “programmed to” refers to one or more processors or cores of the computing element being programmed by a set of software instructions stored in the memory of the computing element to execute the set of functions on target data or data objects stored in the memory.

(120) One should appreciate that the disclosed techniques provide many advantageous technical effects including reduction in latency between a computing device ingesting healthcare data and generating a prediction or recommendation. Latency is reduced through storage of health care data in a memory and in the form of N-grams, which can be computationally analyzed quickly.

(121) Although the present disclosure has been described in detail with reference to particular arrangements and configurations, these example configurations and arrangements may be changed significantly without departing from the scope of the present disclosure. For example, although the present disclosure has been described with reference to particular communication exchanges involving certain network access and protocols, system **10** may be applicable to other exchanges or routing protocols. Moreover, although system **10** has been illustrated with reference to particular elements and operations that facilitate the communication process, these elements, and operations may be replaced by any suitable architecture or process that achieves the intended functionality of system **10**.

(122) Numerous other changes, substitutions, variations, alterations, and modifications may be ascertained to one skilled in the art and it is intended that the present disclosure encompass all such changes, substitutions, variations, alterations, and modifications as falling within the scope of the appended claims. In order to assist the United States Patent and Trademark Office (USPTO) and, additionally, any readers of any patent issued on this application in interpreting the claims appended hereto, Applicant wishes to note that the Applicant: (a) does not intend any of the appended claims to invoke paragraph six (6) or (f) of 35 U.S.C. section 112 as it exists on the date of the filing hereof unless the words “means for” or “step for” are specifically used in the particular claims; and (b) does not intend, by any statement in the specification, to limit this disclosure in any way that is not otherwise reflected in the appended claims.

## Claims

1. A computing device comprising: one or more hardware processors configured to execute program instructions to cause the computing device to: monitor user data corresponding to real-time user activities of a user; update an alarm fatigue level of the user in real-time based on at least the user data, the alarm fatigue level corresponding to a likelihood the user responds to notifications; before generating a first notification of a computer operating system, set an attribute of the first notification of the computer operating system based on at least the alarm fatigue level of the user, wherein the attribute is configured to increase the likelihood the user responds to the first notification to proactively reduce the likelihood of alarm fatigue and a need for a second notification before the alarm fatigue occurs, wherein setting the attribute of the first notification comprises setting one or more of a notification intensity or a notification duration to differentiate the first notification from other notifications to which the user has responded; and generate the first notification with the attribute.
2. The computing device of claim 1, wherein the computing device comprises a personal alarm device.
3. The computing device of claim 1, wherein the computing device comprises a watch.
4. The computing device of claim 1, wherein the one or more hardware processors is further configured to execute the program instructions to cause the computing device to generate the first notification on a personal alarm device, the personal alarm device comprising a watch.
5. The computing device of claim 1, wherein the user data corresponding to the real-time user activities comprises user responses to alerts or notifications.
6. The computing device of claim 1, wherein the user data corresponding to the real-time user activities comprises physiological data of the user, the physiological data comprising one or more physiological parameters relating to a physiological status of the user.
7. The computing device of claim 1, wherein the one or more hardware processors is further

configured to execute the program instructions to cause the computing device to set the attribute of the first notification by setting a notification distribution mode from one mode to another mode including one or more of a visual mode, an auditory mode, or a tactile mode, wherein the notification intensity comprises one or more of a light intensity or an audio intensity based on the notification distribution mode.

8. The computing device of claim 1, wherein the one or more hardware processors is further configured to execute the program instructions to cause the computing device to set the attribute of the first notification by setting one or more of a notification frequency or a notification threshold.

9. The computing device of claim 1, wherein the one or more hardware processors is further configured to execute the program instructions to cause the computing device to set the attribute of the first notification based on at least a subscription level corresponding to a subscription service.

10. The computing device of claim 1, wherein the one or more hardware processors is further configured to execute the program instructions to cause the computing device to set the attribute of the first notification based on at least a subscription level, the subscription level based on at least a location of the computing device or the user.

11. The computing device of claim 1, wherein the one or more hardware processors is further configured to execute the program instructions to cause the computing device to set the attribute of the first notification by deleting the first notification.

12. A computer-implemented method comprising: by one or more computer processors executing program instructions: monitoring user data corresponding to real-time user activities of a user; updating an alarm fatigue level of the user in real-time based on at least the user data, the alarm fatigue level corresponding to a likelihood the user responds to notification; before generating a first notification of a computer operating system, setting an attribute of the first notification of the computer operating system based on at least the alarm fatigue level of the user, wherein the attribute is configured to increase the likelihood the user responds to the first notification to proactively reduce the likelihood of alarm fatigue and a need for a second notification before the alarm fatigue occurs, wherein setting the attribute of the first notification comprises setting one or more of a notification intensity or a notification duration to differentiate the first notification from other notifications to which the user has responded; and generating the first notification with the attribute.

13. The computer-implemented method of claim 12, further comprising generating the first notification on a personal alarm device, the personal alarm device comprising a watch.

14. The computer-implemented method of claim 12, wherein the user data corresponding to the real-time user activities comprises physiological data of the user, the physiological data comprising one or more physiological parameters relating to a physiological status of the user.

15. The computer-implemented method of claim 12, further comprising setting the attribute of the first notification by setting a notification distribution mode from one mode to another mode including one or more of a visual mode, an auditory mode, or a tactile mode, wherein the notification intensity comprises one or more of a light intensity or an audio intensity based on the notification distribution mode.

16. The computer-implemented method of claim 12, further comprising setting the attribute of the first notification by setting one or more of a notification frequency or a notification threshold.

17. The computer-implemented method of claim 12, further comprising setting the attribute of the first notification based on at least a subscription level corresponding to a subscription service.

18. Non-transitory computer-readable media including computer-executable instructions that, when executed by a computing system, cause the computing system to perform operations comprising: monitoring user data corresponding to real-time user activities of a user; updating an alarm fatigue level of the user in real-time based on at least the user data, the alarm fatigue level corresponding to a likelihood the user responds to notifications; before generating a first notification of a computer operating system, setting an attribute of the first notification of the computer operating system

based on at least the alarm fatigue level of the user, wherein the attribute is configured to increase the likelihood the user responds to the first notification to proactively reduce the likelihood of alarm fatigue and a need for a second notification before the alarm fatigue occurs, wherein setting the attribute of the first notification comprises setting one or more of a notification intensity or a notification duration to differentiate the first notification from other notifications to which the user has responded; and generating the first notification with the attribute.

19. The non-transitory computer-readable media of claim 18, wherein the computer-executable instructions, when executed by the computing system, further cause the computing system to perform operations comprising generating the first notification on a personal alarm device, the personal alarm device comprising a watch.

20. The non-transitory computer-readable media of claim 18, wherein the user data corresponding to the real-time user activities comprises physiological data of the user, the physiological data comprising one or more physiological parameters relating to a physiological status of the user.

---